DenseMixerMamba: Residual Mixing for Spectral–Spatial Hyperspectral Image Classification

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***Abstract*—The classification of hyperspectral images (HSIs) is a fundamental but challenging topic in the field of surface monitoring. In recent years, deep learning (DL) models like recurrent neural network (RNN), convolutional neural network (CNN), transformers, and Mamba have demonstrated their remarkable and promising processing capabilities to confront high-dimensional and complicated HSI data. However, how to capture the refined spatial and spectral information from high-dimensional data continues to pose efficiency problems, which are compounded by the increased complexity of deep models. CNNs and transformers employ thousands of parameters, imposing a high computational load and memory consumption. This article addressed the parameter reduction problem by proposing an end-to-end `DenseMixerMamba`network for HSI classification, which converges Mamba, attention mechanisms, and dense architectures. Specifically, channel attention is first introduced to capture valuable channels and stimulate subsequent feature representation quickly. Then, to enhance the flow of information between layers and provide more stable training, a novel encoder based on the Mamba vision mixer (MVM) block and dense connectivity is developed to capture spatial and spectral patterns without increasing computational effort. Furthermore, a dual-branch architecture is designed to support the MVM block, which maintains both local and global information for final representation. Numerous experiments on four real datasets exhibit the more impressive classification accuracy and generalization performance of `DenseMixerMamba`over other state-of-the-art methods.**

***Index Terms*—Attention mechanism, deep learning (DL), dense connection, hyperspectral classification, Mamba.**

Received 13 September 2024; revised 9 February 2025, 22 May 2025, and 20 August 2025; accepted 15 September 2025. Date of publication 6 October 2025; date of current version 30 October 2025. This work was supported in part by the National Natural Science Foundation of China under Grant 42271350 and Grant 42241109, in part by the International Partnership Program of the Chinese Academy of Sciences under Grant 313GJHZ2023066FN, in part by the Research and Development through MICIU/AEI/10.13039/501100011033 under Project PID2023-149880NA-I00, in part by the 2024 Leonardo Grant for Scientific Research and Cultural Creation from the Banco Bilbao Vizcaya Argentaria (BBVA) Foundation (the BBVA Foundation accepts no responsibility for the opinions, statements and contents included in the project and/or the results thereof, which are entirely the responsibility of the authors), and in part by the European Regional Development Fund (ERDF) of the European Union. *(Corresponding authors: Juan M. Haut; Danfeng Hong.)*

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Digital Object Identifier 10.1109/TGRS.2025.3618015

I. Introduction

HYPERSPECTRAL images (HSIs) are valuable observational resources in remote sensing, providing detailed spectral information across numerous contiguous bands, enabling the precise identification and analysis of various materials on the Earth's surface [1]. Hyperspectral classification is a sophisticated and promising task that involves assigning class labels to each pixel using the scene's refined spectral and spatial information. This has been widely applied in precision agriculture, environmental analysis, military surveillance, mineral exploration, and urban surveys. However, this rich spectral information also poses challenges related to the high dimensionality and the inherent complexity of the data, which can complicate traditional classification approaches. Deep learning (DL) has emerged as a promising tool to tackle these challenges, significantly improving the accuracy and efficiency of classification. DL techniques, particularly recurrent neural networks (RNNs), convolutional neural networks (CNNs), and transformers, can learn hierarchical representations of the data, facilitating the extraction of meaningful features that are critical for an accurate classification [2]. Despite the advantages and strengths of these architectures, most rely on the optimization of a large number of parameters to achieve accurate classification, while other architectures, particularly those with sequential processing components, face challenges in fully leveraging compute-parallelization devices. This increases their complexity, directly impacting computational and memory consumption, which is already significant due to the demands of HSI data analysis. A summary of their advantages and limitations in the context of HSI classification is provided below.

A. HSI Classification With RNNs

RNN is a classic architecture for early DL with a cyclic connection structure to propagate information and simultaneously maintain and update dependencies between temporal steps, making it particularly adept at processing data with sequential characteristics. In HSI, each pixel is treated as a spectral sequence due to the sensor's exceptional spectral information acquisition capabilities. RNNs for HSI classification can capture complex patterns and long-range dependencies within these spectral sequences, which is instrumental in distinguishing subtle spectral differences, thereby enhancing classification accuracy. Early RNNs are capable of capturing fundamental sequence dependencies for classification by ana-

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lyzing spectral sequences pixel by pixel. Nevertheless, they would encounter vanishing and exploding gradients inherent to the architecture. To overcome this, more advanced architectures were developed, notably exemplified by long short-term memory (LSTM) networks [3] and gated recurrent units (GRUs) [4]. Both architectures utilize gating mechanisms to regulate information flow and state updates, thereby significantly enhancing the stability and efficacy of RNNs in training on long-sequence data. Nevertheless, the unit structure and recursive mechanism of RNNs do not inherently account for the spatial proximity between pixels, thus limiting their ability to perceive the spatial structure in HSI [5]. Combining CNNs and RNNs can partially address this limitation, improving the overall performance of HSI classification [6].

B. HSI Classification With CNNs

The rapid advancement of CNNs can be largely attributed to the groundbreaking success of AlexNet [7] in image classification, which initially demonstrated the unpredicted potential of deep convolution layers and activation functions for feature mining. Subsequently, as hardware and algorithmic innovations progressed, CNN architectures have become increasingly deeper and more complex, including 2-D (CNN2D) and 3-D (CNN3D) architectures, and multiscale CNNs, significantly furthering the applications of CNNs [2]. In HSI classification, CNNs can automatically capture complex patterns and details in data through multiple layers of convolution and pooling operations without manual intervention, which is crucial for handling the rich spectral information inherent in HSI. The convolution operations enable the simultaneous processing and integration of spectral and spatial information in HSI, thereby preserving and leveraging these two dimensions in feature representation. CNN3Ds particularly prepare the model to capture joint spectral–spatial features more efficiently. In addition, various advanced CNN architectures, such as residual network (ResNet) [8], DenseNet [9], graph CNN (GCNs) [10], and CapsNet [11], have been developed to further improve model performance and training efficiency. These architectures, with deeper layers, more complex network structures, and innovative connection strategies, significantly enhance the ability to learn the intricate patterns present in HSI. However, CNNs, which extract features via local convolutional kernels, tend to lack the capability to capture global features effectively, resulting in poor performance in HSI classification tasks. Moreover, each pixel in an HSI encompasses a spectral sequence, yet the localized receptive fields of CNNs hinder their ability to capture the global dependencies within these spectral sequences, i.e., global information about the entire spectral sequence cannot be integrated.

C. HSI Classification With Transformers

In recent years, transformers have spurred a paradigm shift in DL, becoming a highly sought-after architecture due to its innovative design and powerful feature extraction capabilities. Through a multihead self-attention mechanism and positional encoding, transformers compute the similarity weights between all elements in an input sequence to efficiently capture complex patterns and long-range dependencies, setting a new benchmark across various tasks [12]. Unlike traditional RNNs, the transformer can process sequence data in parallel, significantly enhancing training efficiency. The vision transformer (ViT) is specifically optimized for computer vision tasks by partitioning images into fixed-size patches, flattening them, and processing them as a sequence, thus improving its ability to capture spatial features [13]. In terms of HSI classification, spectral transformers [14], spectral–spatial transformers [15], multiscale transformers [16], and CNN-transformers hybrid models [17] are devised to capture multilevel spatial and spectral information in HSI, comprehensively improving classification performance. However, this architecture still suffers from unavoidable shortcomings. Specifically, the self-attention mechanism of transformers requires the computation of interactions between all elements in a sequence, leading to computational and memory complexity that scales quadratically with the sequence length. Consequently, transformers are significantly less efficient in applications dealing with high-dimensional contexts and constrained computational resources, much less in real-time processing tasks.

D. New Era and Challenges for HSI Classification

Essentially, sequence models such as RNNs are architectures similar to state space models (SSMs), which utilize state and output equations to achieve mathematical modeling, providing a more powerful framework for describing complex dynamic systems [18]. More excitingly, recent research has revealed a wealth of promising applications for SSMs, in which Mamba is the most groundbreaking achievement, garnering widespread attention [19]. In Mamba, the state space is selectively designed to achieve the modeling capabilities of transformers while enabling linear scaling with sequence length, overcoming the limitations of transformers. Extensive experiments have shown that Mamba is five times faster than transformers, capable of handling sequences up to a million in length with comparable or even superior accuracy [19]. Furthermore, vision Mamba (ViM) [20] and visual Mamba (VMamba) [21] have successfully tackled the vision-related task, where the former introduces a Mamba-ViT backbone and the latter develops groundbreaking visual state-space (VSS) blocks to process the information contained in scenes.

Despite these advances, existing Mamba models are typically designed specifically for time series data, facing limitations in HSI tasks where effective utilization of high-dimensional spectral and spatial information is crucial. The specifications are as follows.

1) Each pixel contains hundreds of contiguous spectral bands covering a broad range of the electromagnetic spectrum. These bands are susceptible to interference from uncontrolled atmospheric and environmental factors, as well as external noise. Consequently, this can lead to spectral distortions, increased data variability, and the presence of redundant or superfluous information.

2) The spatial–spectral structure of HSI data differs significantly from the 1-D sequence data normally processed by

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these architectures. Indeed, spatial and spectral dimensions are intrinsically linked, as each pixel encodes a unique spectral signature influenced by its spatial context. This interdependence implies that spatial features can reinforce spectral coherence, making joint spectral–spatial processing essential for accurately distinguishing materials and minimizing classification errors.

3) HSI data do not completely have order dependence, but also have local similarity attributes, including spatial and spectral. Furthermore, despite being models with linear complexity, the increasing complexity of their structures can lead to significant overfitting problems and increasing computational complexity.

To overcome these challenges, this work proposes the novel end-to-end `DenseMixerMamba` network for HSI classification, built upon three core principles: i) extraction of salient features; ii) integrated spectral–spatial data processing; and iii) efficient reuse of features. These principles are driven by the need to dynamically select the most relevant information—thus, mitigating the impact of noise and spectral redundancy— and the combined analysis of interdependent features. Moreover, by ensuring the effective reuse and propagation of features across multiple levels of the architecture, the model minimizes the loss of critical information while promoting the learning of complex patterns. This strategy addresses the lack of ordered dependencies between bands, enabling dynamic adaptation to local variations, reducing overfitting, and improving generalization.

As a result, the proposed `DenseMixerMamba` includes three phases, i.e., initial embedding, dense Mamba mixer encoder, and classification tail, which elaborately leverages the strengths of Mamba, attention mechanisms, and dense architectures to achieve comprehensive feature integration efficiently. The main contributions are as follows.

1) To fast capture salient channels, `DenseMixerMamba` introduces spectral attention during the embedding stage, acting as a spectral purifier that dynamically highlights informative channels and suppresses noninformative data. Unlike conventional designs applied posthoc, the extracted attention features are further refined via stem layers to enhance deeper feature extraction in subsequent layers. In fact, this step improves both the quality of the representation and the stability of the training while reducing the computational efforts devoted to extracting useful patterns in subsequent stages of the network.
2) A novel encoder based on Mamba vision mixer (MVM) block and dense connections is devised to capture spatial and spectral patterns without increasing computation. This dense connectivity facilitates enhanced information flow between layers and improves the preservation and reinforcement of feature maps across the network while mitigating the risk of vanishing gradients, leading to more stable and efficient training. Two different dense connections are elaborated, i.e., residual feature summation (+) and feature concatenation (.).
3) Within each MVM block, spectral and spatial paths are implemented to extract the respective features. These are combined with the original input at the end of the block, resulting in a joint representation while preserving critical information and minimizing content loss. Delving into each path, a new dual-branch architecture is developed to support the MVM block, where the first branch is composed of regular convolution, activation, and SSM modules, whilst the second one implements identical convolution and activation layers without the SSM. This design enables the extraction of both local, specific features and more general, global patterns, facilitating the learning of complementary representations. In addition, it acts as a form of regularization, preventing overfitting to specific patterns and minimizing the premature loss of critical information.

The proposed design reduces the number of parameters to be trained by the network, in particular the `DenseMixerMamba` (+). This results in a lower use of computational and memory resources, allowing the `DenseMixerMamba` (+) to be used more efficiently in environments with limited computational and memory capacity.

Extensive experiments demonstrate that `DenseMixerMamba` achieves extremely satisfactory accuracy and throughput, while reducing the number of parameters, where densely connected MVM blocks produce richer feature representations and better generalization capability.

The rest of the manuscript is organized as follows. Section II provides a comprehensive review of related works, delving into its benefits, as well as its limitations and the gaps that we intend to fill with our `DenseMixerMamba` model. Section III describes the details of the proposed method. Section IV presents the classification results on four real HSI datasets and offers an in-depth analysis of the method's performance. Finally, Section V summarizes the work and gives a conclusion.

II. Related Works: From RNNs to Mamba

Hyperspectral remote sensing images (HSIs) classification is a challenging task due to the high dimensionality and spectral variability of the data [22], which often lies in a nonlinear complex feature space, and it is further exacerbated by the paucity of labeled samples [23]. Thus, significant research efforts have been dedicated to developing improved techniques for effective feature extraction and interpretation, undergoing significant advancements with the adoption of DL models [2], [24], [25]. This work details the most relevant deep models provided by the scientific community that have served as the foundational basis for the proposed method. To this end, a brief review of DL mechanisms for HSI classification is presented, commencing with RNNs, and progressing to the latest advancements in visual transformers and SSMs.

A. Early DL Approaches: RNNs in HSI Classification

Following traditional pattern recognition and standard machine learning models [26], early attempts at DL for HSI classification analyzed the hyperspectral scene by means of its spectral features, processing the spectral content of the N-dimensional pixel vector in isolation to infer the corresponding

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land cover or land use category (with N indicating the number of bands) [27], [28], [29], [30].

In this context, RNNs are remarkable for their ability to capture temporal dependencies in sequential data, particularly short-term dependencies. This is achieved by incorporating information from previous inputs through recurrent connections between neural activations. In the HSI classification context, when considering a $M \times N$ scene, where N is the number of spectral bands and $M = HW$ collects the spatial height and widths dimensions, the hyperspectral pixel is treated as the spectral sequence $\mathbf{x}_i = \{x_{i,1}, \ldots, x_{i,N}\}$, introducing the band values $x_{i,t}$ into the network as a sequence of inputs over time steps $t = 1, \ldots, N$. The spectral feature $x_{i,t}$ is then processed by the hidden layer (also known as memory cell or RNN unit), which computes both the corresponding output $\mathbf{y_t} \in \mathbb{R}^{N_y}$ and a hidden state $\mathbf{h}_t \in \mathbb{R}^{N_h}$

$$\begin{aligned} \mathbf{y}_t &= \phi_y\left(\mathbf{W}_y \mathbf{h}_t + \mathbf{b}_y\right) \\ \mathbf{h}_t &= \begin{cases} 0, & \text{if } t = 0 \\ \phi_h\left(\mathbf{W}_x x_t + \mathbf{U}_h \mathbf{h}_{t-1} + \mathbf{b}_h\right), & \text{if } t \neq 0 \end{cases} \end{aligned} \tag{1}$$

where $\mathbf{W}_x \in \mathbb{R}^{N_h}$, $\mathbf{U}_h \in \mathbb{R}^{N_h \times N_h}$, and $\mathbf{W}_y \in \mathbb{R}^{N_h \times N_y}$ weight the input-to-hidden, hidden-to-hidden, and hidden-to-output connections, respectively, with N_h and N_y indicating the hidden and output spaces; $b_h \in \mathbb{R}^{N_h}$ and $b_y \in \mathbb{R}^{N_y}$ are the hidden and output biases; and ϕ_h and ϕ_y are activation functions, where the former is usually performed by sigmoid, hyperbolic tangent or rectified linear unit (ReLU) activation functions, and the latter by the softmax function. It is noteworthy that these weights and biases are shared along the entire sequence of size N, whilst the hidden state $\mathbf{h}_t$ is fed back to the memory cell for the computation of the following feature $x_{i,t+1}$, thus retaining information from previous time steps and modeling the underlying patterns of the input spectral sequence over the channels.

Naive or vanilla RNNs have been previously explored in HSI classification [31]. For instance, Wu and Prasad [32] combined the outstanding feature extraction capability exhibited by CNNs with the sequence analysis of RNNs (the so-called CRNN [33]). Also, Wu and Prasad [34] explored the CRNN within a pseudolabeling scheme for semi-supervised learning.

Despite their simplicity, or precisely because of it, vanilla RNNs are not suitable for high-dimensional data. Although they can be unrolled regardless of the magnitude N (i.e., without the trainable weights and biases being modified, so there is no need to store N copies of them), the accumulation of unrolled weights along the time steps is equal to raising them to the exponent N. As a result, the gradients can explode when the weights are larger than 1, or vanish when the weights are smaller than 1. This hinders their ability to learn from long spectral sequences, a critical aspect of HSI data. To overcome this limitation, different mechanisms have been designed to control the information flow within the memory cell, such as the GRU and its bidirectional implementation (BiGRU), the simple recurrent unit (SRU) [35], and the LSTM unit.

In particular, the LSTM shows remarkable efficiency in capturing long-term dependencies. Its structure is equipped with a cell state and a three-gate mechanism, which comprises forget, input, and output gates to control the data flow, selectively storing and retrieving information over long sequences. As a result, these models withstand the HSI dimensionality curse far better. For instance Zhou et al. [36] combined the standard spectral-based LSTM—where the N bands comprising the pixel $\mathbf{x}_i$ are introduced into the network one by one to learn the spectral feature—and a spatial-based LSTM—where the target pixel is considered as the sequence of its P neighboring pixels, preprocessed by PCA and introduced one by one into the network to learn spatial relationships—by means of a decision fusion strategy. Prior feature extraction has been frequently employed to enhance the classification outcomes of recurrent networks, exploiting the unmatched capacity of CNNs to learn abstract hierarchical features, from low-level edges to high-level patterns. For instance, in [32] a CLSTM was presented, in which the CNN1D extracted spectral features that were then processed by the LSTM, whilst Seydar et al. [37] used a two-layer CNN3D to extract the spectral–spatial features.

Notwithstanding their capacity to learn high-level features directly from raw data—reaching impressive results—recurrent models still face important challenges in achieving both effectiveness and efficiency in HSI classification. Despite the development of prior feature extraction procedures, standard LSTMs often struggle with the analysis of complex spatial contextual information, as these models are primarily focused on spectral information. Moreover, LSTMs typically require a large amount of training data to achieve optimal performance, which is a significant obstacle to HSI classification due to the scarcity of labeled samples. In addition, the inherently sequential nature of these models makes them difficult to parallelize, requiring significant computational resources to conduct the training procedure, which is highly time-consuming due to the high dimensionality of the data.

B. CNNs and Attention: Toward the Transformer Revolution

CNNs and the subsequent introduction of attention mechanisms have significantly advanced the HSI processing field, overcoming the limitations of sequential models. Indeed, the convolution kernel is inherently designed to process multidimensional data, and is implemented as a tensor comprising a set of F filters applied either to a C channel window (1-D), a $K \times K$ spatial window (2-D), or both (3-D). Within the hierarchical stack of convolution layers, each layer convolves its kernel over its input feature maps, producing local connectivity, and conveys the results to the subsequent layer. This mechanism rigidly encodes a certain invariance in the analysis and introduces an inductive bias, whereby entries that are close to each other are more closely related than those that are further apart. In this regard, although the fixed size of its receptive field may limit its ability to capture long-range dependencies [38], the convolution kernel has proved to be a great feature extractor, preserving the spectral–spatial structure of the inputs and learning hierarchical features directly from raw data, from low-level ones to high-level patterns. As a result, CNNs are exceptionally suited for capturing spectral–spatial relationships

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between hyperspectral pixels. The available literature on these models is extensive [39], [40].

Although CNN has dominated the state-of-the-art in image processing [41], [42], it is not without certain limitations. In this sense, CNNs focus on local connectivity and may be missing global context due to the localized nature of the kernel. Moreover, the same convolution kernel is applied—sliding with a stride—to the entire input, which may not be optimal for all regions of the data. In addition, CNNs fail to capture long-range dependencies, as the feature data representation is captured in a limited receptive field, and their layer-wise structure lacks mechanisms to account for sequential dependencies. To overcome these limitations without waiving the great feature extraction capacity of CNNs, a variety of attention mechanisms have been designed in recent years.

Attention mechanisms were originally designed to be applied in sequence-to-sequence (Seq2Seq) models for neural machine translation tasks. The encoder–decoder architecture of the Seq2Seq (with an RNN-based backbone) is intended to produce a variable-length sequence of tokens from a variable-length sequence of input data. The procedure starts with the encoding of the variable-length input into a fixed-size embedding representation, which is finally decoded into the variable-length output. In this context, the attention mechanism is based on a weighted combination of features by learning an attention mask. This enables the model to focus on the most relevant parts of the data whilst masking the less pertinent ones regardless of their position, i.e., to pay attention to certain input features instead of processing all information in a uniform manner.

With the advent of multibranch, nonlocal models [43], ResNets, and the positive impact of the squeeze-and-excitation network (SE-net), CNN models started to integrate early additive and multiplicative attention strategies into their architectures as plug-and-play modules, which applied masks to both the spectral and spatial dimensions of the data. For instance, SE-net applies attention to the spectral dimensions of the data. The feature maps are squeezed into a vector which encodes the global information across all spatial locations for each channel. Other mechanisms, such as convolutional block attention module (CBAM) [44], attention dropout convolutional module (ADCM) [45], or dual attention network (DAN) [46], also combine spectral and spatial attention, and have been widely used in HSI classification [47].

By emphasizing the salient features, these solutions dynamically adjust the focus across the entire input, offering a more interpretable model and making their comprehension and analysis easier. Nevertheless, although attention mechanisms enhance the processing of long-range dependencies and the acquisition of certain global contexts, these modules still operate largely within a local context, neglecting complex spectral–spatial relationships and sequential dependencies. Furthermore, integration of attention modules into existing network architectures may be challenging, introducing additional computational complexity and memory requirements, and requiring careful design and expertise on the task and the data to process.

C. Transformers: A Paradigm Shift in HSI Analysis

Within this framework, the transformer network and its self-attention mechanism (also known as intra-attention) bring a new revolution within DL [12].

1) Self-Attention: Contrary to standard attention mechanisms, which relate different parts between the input-encoded and the output-decoded sequences through context vectors, the self-attention mechanism models the significance of each element (token) of the input sequence, i.e., it generates a context for each token by considering just the sequence itself.

In Fig. 1(a), let consider $\mathbf{X}$ as a sequence of size $M \times N$, where M is the number of elements and N denotes the dimension of each element, i.e., $\mathbf{x}_i = [x_{i,1}, \ldots x_{i,N}]$, $i = 1, \ldots, M$. The self-attention mechanism attempts to incorporate richer contextual information into each $\mathbf{x}_i$. For this purpose, it projects $\mathbf{X}$ into query $\mathbf{Q} \in \mathbb{R}^{M \times D_q}$, key $\mathbf{K} \in \mathbb{R}^{M \times D_k}$, and value $\mathbf{V} \in \mathbb{R}^{M \times D_v}$ components, following the linear transformations of

$$\begin{aligned} \mathbf{Q} &= \left[\mathbf{q}_1, \ldots, \mathbf{q}_M\right], \quad \text{with } \mathbf{q}_i = \mathbf{W}_q \mathbf{x}_i \\ \mathbf{K} &= \left[\mathbf{k}_1, \ldots, \mathbf{k}_M\right], \quad \text{with } \mathbf{k}_i = \mathbf{W}_k \mathbf{x}_i \\ \mathbf{V} &= \left[\mathbf{v}_1, \ldots, \mathbf{v}_M\right], \quad \text{with } \mathbf{v}_i = \mathbf{W}_v \mathbf{x}_i \end{aligned} \tag{2}$$

where $\mathbf{W}_q \in \mathbb{R}^{D_q \times N}$, $\mathbf{W}_k \in \mathbb{R}^{D_k \times N}$, and $\mathbf{W}_v \in \mathbb{R}^{D_v \times N}$ are the transformation matrices from the input space to the query, key, and value space, respectively. For simplicity, the matrix dimensions usually coincide, i.e., $D_q = D_k = D_v \ll N$, particularly when the scaled dot-product attention is conducted (imposing $D_q = D_k$).

It is noteworthy that: $\mathbf{q}_i$ stands for the information the ith token is seeking from the other tokens, i.e., $\mathbf{Q}$ contains feature vectors; $\mathbf{k}_i$ comprises those features of the token that could impact the other tokens, i.e., $\mathbf{K}$ can be seen as a collection of masks; and $\mathbf{v}_i$ is the actual content of the ith token, thus, $\mathbf{V}$ contains the tokens. In light of this, the self-attention mechanism computes scores between query and key components through the dot product, measuring the similarity between each $\mathbf{q}_i$ and $\mathbf{k}_j$, $\forall i, j \in [1, M]$. Particularly, large (positive) scores indicates that $\mathbf{q}_i$ and $\mathbf{k}_j$ are closely aligned, which means that $\mathbf{k}_j$ is highly relevant for $\mathbf{q}_i$. Conversely, small (negative) scores suggest an inferior alignment between $\mathbf{q}_i$ and $\mathbf{k}_j$, i.e., low relevance. As a result, the obtained raw attention scores describe the pairwise relevance between all elements in the sequence. These scores are then scaled by D_k for the sake of the stability when processing data with large dimensions and normalized using the softmax to obtain the matrix of attention weights $\boldsymbol{\alpha} \in \mathbb{R}^{M \times M}$. Finally, attention weights are applied to $\mathbf{V}$ to obtain $\mathbf{Z} \in \mathbb{R}^{M \times D_v}$, a weighted representation determined by how much attention the tokens are paying to the other elements in the sequence

$$\begin{aligned} \alpha_{i,j} &= \text{Softmax}\left(\frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{D_k}}\right)_{i,j} = \frac{\exp\left(\frac{\mathbf{q}_i \cdot \mathbf{k}_j}{\sqrt{D_k}}\right)}{\sum_{j=1}^{M} \exp\left(\frac{\mathbf{q}_i \cdot \mathbf{k}_j}{\sqrt{D_k}}\right)} \\ \mathbf{Z} &= \boldsymbol{\alpha}\mathbf{V}. \end{aligned} \tag{3}$$

2) Multihead Attention: In the basic self-attention mechanism, a single attention head may focus on specific aspects of the sequence. As a result, the model is limited to capturing a single type of relationship at a time. To focus on different

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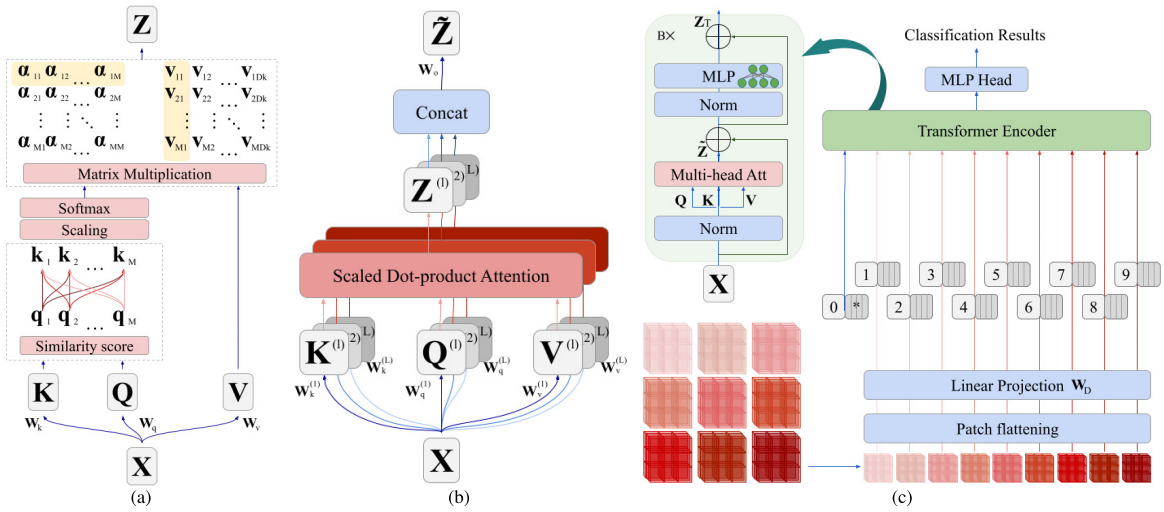


Fig. 1. Graphical representation of the original transformer network components: (a) self-attention mechanism; (b) multihead attention module, and (c) ViT architecture with transformer block.

kinds of information simultaneously, the transformer extends the attention to multiple heads. Considering L heads, for each attention head separate projections of $\mathbf{X}$ are conducted through $\mathbf{W}_q^{(l)}$, $\mathbf{W}_k^{(l)}$, and $\mathbf{W}_v^{(l)}$ to generate its own $\mathbf{Q}^{(l)}$, $\mathbf{K}^{(l)}$, and $\mathbf{V}^{(l)}$. This reduces the dimension considerably, setting $D_k = N/L$.

Fig. 1(b) provides the multihead scheme, where each head performs (3) on its data, producing $\mathbf{Z}^{(l)}$. Each output is then concatenated in a tensor of size $M \times (L \cdot D_v)$, and linearly projected through output weights $\mathbf{W}_o$ into the final $\tilde{\mathbf{Z}}$ values.

3) Vision Transformer: The multihead attention is encapsulated in a transformer block, the architecture of which may vary according to the type of network. In this regard, the ViT has revolutionized image processing [13]. Given an input image of size $H \times W \times N$, the ViT crops and flattens it into $M = \mathrm{HW}/P^2$ patch vectors of size $(P^2 \cdot N)$, where $P \times P$ is the patch size. Patch vectors are then projected into the latent space through a trainable linear projection with parameters $\mathbf{W}_p \in \mathbb{R}^{D \times (P^2 \cdot N)}$.

To preserve positional information, positional embeddings are appended to the resultant patch embeddings of size D. Furthermore, an extra-learnable [CLS] token is added to capture the global context of the entire image. Obtained embeddings $\mathbf{X} \in \mathbb{R}^{M \times D}$ are fed to the encoder, which comprises B transformer blocks equipped with the multihead attention mechanism. Fig. 1(c) illustrates the process.

A wide variety of transformers have been adapted for HSI classification due to their comprehensive understanding of both spectral and spatial feature patterns.

He et al. [15] designed a spectral–spatial transformer, where a CNN is used to extract the spatial features, and a densely connected transformer captures the sequential spectra relationships. Qing et al. [14] combined the transformer network with spectral attention to enhance the initial feature representation. Notwithstanding their impressive performance, the transformer network faces several limitations and challenges in relation to feature extraction.

In this context, Hong et al. [48] proposed the SpectralFormer backbone, which focuses especially on the extraction of spectral information to overcome limitations in modeling locally detailed spectral discrepancies. Roy et al. [49] developed the morphological transformer with an improved multihead patch attention module to obtain discriminative spectral–spatial features. Zhou et al. [50] coped with the transformers over-fitting issue by including contrastive learning. Sun et al. [17] proposed the spectral–spatial feature tokenization transformer, comprising 3-D and 2-D convolution layers for feature extraction and introducing a Gaussian weighted feature tokenizer for feature transformation. To cope with the over-dispersion exhibited by features extracted through the multihead self-attention mechanism, Mei et al. [51] introduced a grouped pixel embedding module into each transformer block. Also, to reduce the high computation burden, Zhao et al. [52] proposed the group-wise separable convolutional transformer to enhance the local spectral–spatial information whilst lightening the transformer architecture. Furthermore, Roy et al. [53] designed two lightweight self-attention modules, namely, the channel lightweight multihead self-attention module and position lightweight multihead self-attention module, respectively, to reduce memory consumption and computational requirements.

Despite these improvements, transformers still suffer some limitations related to translation invariance, inductive biases, or overfitting, in addition to their high computational demand due to their complex architecture. Furthermore, the maximum input sequence length (or context window) poses a significant obstacle as it determines the number of elements, i.e., the number of tokens from the original data that the transformer model can process. It is quadratically related to the number of parameters in the model, increasing the computational overhead of the model and impacting its performance when obtaining more nuanced and coherent outputs [54].

D. Mamba: Expanding Opportunities for HSI Analysis

The high computational complexity of sophisticated networks such as transformers is prohibitive when processing high-dimensional HSI. In this regard, SSMs in general and Mamba in particular are gaining growing attention to cope with long and complex sequences [19].

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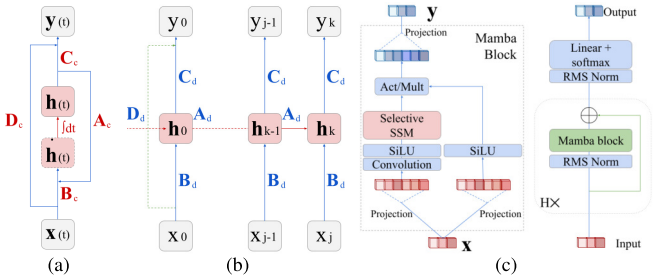


Fig. 2. Graphical overview of SSM in (a) continuous-time and (b) discrete-recurrent representations. (c) Structure of Mamba block.

1) State Space Model: SSM has been recently introduced into DL. It provides a mathematical description of a system by analyzing its internal state, understanding its evolution over time in response to inputs, and modeling the generated output. Following the linear Gaussian SSM formulation (inspired by Kalman filters), the SSM is the continuous-time differential equation [see Fig. 2(a)]

$$\begin{aligned}\dot{\mathbf{h}}(t) &= \frac{d\mathbf{h}(t)}{\mathrm{dt}} = \mathbf{A}_c\mathbf{h}(t-1) + \mathbf{B}_c x(t)\\ \mathbf{y}(t) &= \mathbf{C}_c\mathbf{h}(t)\,[+\mathbf{D}_c x(t)]\end{aligned} \tag{4}$$

where $\mathbf{h}(t) \in \mathbb{R}^D$ is a set of state variables relating the input data $x(t) \in \mathbb{R}$ observed at time t, with the output $y(t) \in \mathbf{R}$, $\mathbf{A}_c \in \mathbb{R}^{D\times D}$ is the state matrix and controls the impact of the previous latent state on the current state, $\mathbf{B}_c \in \mathbb{R}^{D\times 1}$ is the input matrix and defines how the input data influences on the current state, $\mathbf{C}_c \in \mathbb{R}^{1\times D}$ is the output matrix and controls the impact of the current state on the output, and $\mathbf{D}_c \in \mathbb{R}^{1\times 1}$ is the direct transition matrix that controls the skip connection between the input and the output (which will be omitted for the sake of simplicity). Subindex c indicates "continuous" parameters. D defines the latent space dimension.

This (4) evokes to some extent (1). Indeed, to convert the continuous-time model into the discrete-time model (5), the system behavior must be approximated at discrete time steps $t = j\Delta$, where j indicates the discrete time step and Δ represents the time step size for discretization

$$\begin{aligned}\mathbf{h}_j &= \mathbf{A}_d\mathbf{h}_{j-1} + \mathbf{B}_d x_j\\ y_k &= \mathbf{C}_d\mathbf{h}_j.\end{aligned} \tag{5}$$

Different techniques can be applied. Considering the *zero-order hold*, the discrete counterparts $\mathbf{A}_d$, $\mathbf{B}_d$, $\mathbf{C}_d$, and $\mathbf{D}_d$ are obtained through the closed expressions $\mathbf{A}_d = \exp(\Delta\mathbf{A}_c)$, $\mathbf{B}_d = (\Delta\mathbf{A}_c)^{-1}(\exp(\Delta\mathbf{A}_c) - \mathbf{I}) \cdot \Delta\mathbf{B}_c$, $\mathbf{C}_d = \mathbf{C}_c$, and $\mathbf{D}_d = \mathbf{D}_c$, where Δ is the learnable step size.

Fig. 2(b) provides the discrete recurrent representation. Indeed, unrolling $\mathbf{h}_j$ in (5) yields the scan operations

$$\begin{aligned}\mathbf{h}_0 &= \mathbf{B}_d x_0\\ \mathbf{h}_1 &= \mathbf{A}_d\mathbf{B}_d x_0 + \mathbf{B}_d x_1\\ &\cdots\\ \mathbf{h}_j &= \mathbf{A}_d^j\mathbf{B}_d x_0 + \mathbf{A}_d^{j-1}\mathbf{B}_d x_1 + \ldots + \mathbf{A}_d\mathbf{B}_d x_{j-1} + \mathbf{B}_d x_j\end{aligned}$$

and thus unrolling y_j yields

$$\begin{aligned}y_0 &= \mathbf{C}_d\mathbf{B}_d\mathbf{x}_0\\ y_1 &= \mathbf{C}_d\mathbf{A}_d\mathbf{B}_d\mathbf{x}_0 + \mathbf{C}_d\mathbf{B}_d\mathbf{x}_1\\ &\cdots\\ y_j &= \mathbf{C}_d\mathbf{A}_d^j\mathbf{B}_d x_0 + \mathbf{C}_d\mathbf{A}_d^{j-1}\mathbf{B}_d x_1 + \ldots + \mathbf{C}_d\mathbf{B}_d x_j.\end{aligned}$$

Accordingly, coefficients can be expressed through the parallelizable kernel $\mathbf{K} = (\mathbf{C}_d, \mathbf{A}_d^j, \mathbf{B}_d)_{j\in[0,M-1]} \in \mathbb{R}^M$ covering the length of the input sequence (M), and thus, (5) is reformulated as the convolution (6). This convolution kernel is calculated by fast Fourier transform (FFT)

$$y = x * \mathbf{K},\ \mathbf{K} = \left(\mathbf{C}_d\mathbf{B}_d, \ldots, \mathbf{C}_d\mathbf{A}_d^{M-1}\mathbf{B}_d\right). \tag{6}$$

It is noteworthy that, in the context of N-dimensional data $\mathbf{x} \in \mathbb{R}^{M\times N}$, N independent SSM blocks map each channel of $\mathbf{x}$ to the corresponding output channel in $\mathbf{y} \in \mathbb{R}^{M\times N}$.

2) Structured State Space Sequence (S4) Model: Contrary to transformers, which dynamically adapt their attention based on the input sequence, parameters $\mathbf{A}_d$, $\mathbf{B}_d$, $\mathbf{C}_d$, and $\mathbf{D}_d$ exhibit a linear time-invariant (LTI) property, i.e., they stay constant for every token at any temporal shift (6). As a result, SSM cannot perform content-aware processing, limiting their ability to model long-range dependencies. To cope with this, S4 focuses on $\mathbf{A}$, using initialization methods such as HiPPO [55], the projection operators of which inherently include a memory mechanism to determine how much of the past information (old tokens) is retained.

3) Selective Structured SSM (Mamba): To address content-awareness, Mamba dynamically filters and processes information based on the content whilst providing efficient memory management, providing a mechanism to selectively remember or ignore some parts of the input. In particular, two main contributions are provided: i) a selective scan algorithm for information filtering and ii) a hardware-aware algorithm for efficient computation, which combined provide the selective SSM (S6) block.

Regarding the selective scan algorithm, before discretization, Mamba reformulates its parameters as functions of the input, particularly linear parameterized projections are conducted as $\mathbf{B}_c = \mathrm{Linear}_D(\mathbf{x}) \in \mathbb{R}^{M\times D}$ $\mathbf{C}_c = \mathrm{Linear}_D(\mathbf{x}) \in \mathbb{R}^{M\times D}$, and $\Delta = \mathrm{Softplus}(\mathrm{Parameter} + \mathrm{Broadcast}_N(\mathrm{Linear}_1(\mathbf{x}))) \in \mathbb{R}^{M\times N}$, thus, for every input token, different $\mathbf{B}_c$, $\mathbf{C}_c$, and Δ are dynamically computed, selectively highlighting the relevant information to keep in the hidden state. This limitation precludes the use of parallelizable convolutional representations, restricting the approach to recurrent representations depicted by Fig. 2(b). These sequential scan operations hinder the utilization of GPU data parallelism, indeed. To overcome this limitation, Mamba implements a parallel scan algorithm by exploiting the associative property.

Furthermore, to avoid inefficient data transmissions between SRAM and DRAM, Mamba introduces the hardware-aware computation through a *kernel fusion* strategy, which conducts the computations of the discretization, the scan, and the final multiplication by $\mathbf{C}$ in SRAM, preventing the writing of intermediate results in DRAM. In this sense, intermediate states are not stored and have to be *recomputed* during the backward step by using SRAM again, which is more efficient than reading them from DRAM.

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Finally, Mamba introduces the S6 (comprising SSM, HiPPO, parallel selective scan, and hardware-aware structure) into its architecture as building blocks, following Fig. 2(c).

Mamba has shown notable results in computer vision tasks. As a result, it has been rapidly adopted for HSI processing. Pan et al. [56] proposed the MambaLG, which introduces a local and global spatial modeling module and a short- and long-range spectral dynamic perception module to cope with the challenges of receptive fields and quadratic complexity. Li et al. [57] designed the MambaHSI with a spatial and spectral Mamba block and the corresponding feature fusion module, which integrates spatial and spectral information from HSI. Yang et al. [58] introduced the HSIMamba, which incorporates bidirectional state space modules into the Mamba block and a CNN tail to conduct spectral–spatial analysis. Huang et al. [59] developed the SS-Mamba, with separate spectral and spatial token generators that feed the corresponding stack of spectral and spatial Mamba blocks, the outputs of them fused to conduct HSI classification. On the contrary, He et al. [60] implemented the 3DSS-Mamba with a spectral–spatial token generator to obtain 3-D tokens comprising spectral–spatial information. Furthermore, it introduces a selective scanning mechanism into 3-D space to form 3-D spectral–spatial Mamba blocks. Sheng et al. [61] presented the dual-Mamba as a dual-branch network composed of Mamba and CNN blocks, each one splitting into spectral and spatial subbranches, to extract comprehensive information. In particular, the Mamba branch introduces a spectral–spatial cross-attention strategy, whilst the CNN branch implements a lightweight spectral–spatial residual convolution module, enhancing the extraction of global and local relationships. Zhou et al. [62] introduced the Mamba-in-Mamba (MiM) model, where several centralized Mamba-cross-scan (MCS) convert the HSI data into different sequential data that are processed by tokenized Mamba encoders. These enhance the feature generation. Finally, Wang et al. [63] introduced the LE-Mamba with local enhanced S6 and central region spectral S6 blocks to improve HSI classification.

Although the aforementioned models attempt to address feature utilization issues by modifying the scanning strategy or constructing spectral and spatial Mamba blocks, they introduce significant latency. Moreover, the increased complexity may lead to the risk of overfitting, and might not always result in improved accuracy. Therefore, efficiently integrating spatial–spectral information when using Mamba for HSI classification is crucial.

III. Methodology

This section elaborates on the novel design of the proposed end-to-end `DenseMixerMamba` model for HSI classification. Two main contributions are provided. On the one hand, the proposed network integrates vision-friendly MVM blocks within the building blocks, thereby enhancing image analysis capabilities and improving overall accuracy (OA). On the other hand, dense connections between MVM blocks have been implemented to leverage the strengths of dense connectivity within deep neural networks, facilitating better feature reuse, improved gradient flow, and increased network capacity. Two versions of the dense connection have been developed, i.e., feature concatenation and residual feature summation. Furthermore, the proposed `DenseMixerMamba` processes the HSI image to capture long-range interactions in both the spectral and spatial domains, and then processes it to perform the classification task.

The core components of the `DenseMixerMamba` network are clarified to provide a comprehensive understanding of the design choices and implementation details, offering insights into the innovative aspects and advantages of the proposed network architecture. Fig. 3 illustrates the architecture of the model is composed of three main components: i) initial embedding; ii) dense MVM encoder; and iii) classification tail.

A. Initial Embedding

In Fig. 3, $\mathbf{X} \in \mathbb{R}^{H\times W\times B}$ is the input HSI with $H \times W$ spatial dimensions and B spectral bands. First, an attentional process on the spectral dimension is conducted to highlight the salient and more descriptive channels whilst retaining the spatial information. The attention mask $\tilde{\mathbf{m}} \in \mathbb{R}^{B}$ is obtained and refined through two fully connected (FC) layers, and applied along the input channels by $\mathbf{Xm} = \mathbf{m} \cdot \mathbf{X}$.

Masked input $\mathbf{Xm} \in \mathbb{R}^{H\times W\times B}$ is then processed to obtain the embedding representation $\mathbf{Xe} \in \mathbb{R}^{H\times W\times N}$, where a $N@1 \times 1$ Conv2D projects the data into the N-dimensional embedding space, the Group Normalization divides the N channels into groups and normalizes the features within each group, and the ReLU activation function captures the nonlinearity of data.

B. Dense MVM Encoder

The encoder comprises three densely connected MVM blocks interleaved with average pooling layers, strategically enhancing feature utilization while simultaneously reducing spatial dimensions to alleviate the computational burden. On the one hand, dense connections allow for more comprehensive learning and a more detailed and nuanced understanding of the input data by ensuring that information from earlier blocks is continuously propagated and integrated into later blocks. On the other hand, the integration of average pooling layers between the MVM blocks is essential for managing the spatial dimensions of the feature maps, down-sampling them by a power of 2 factor within each pooling operation. In this context, let denote $\mathbf{Xe}^{(l-1)} \in \mathbb{R}^{H'\times W'\times N'}$ and $\mathbf{Xe}^{(l)} \in \mathbb{R}^{H^*\times W^*\times N^*}$ as the input and output feature maps of the lth pair of MM-block and pooling layer,[1] respectively. Two approaches for dense connectivity have been investigated: *feature concatenation* and *feature summation*, following the residual model. Feature concatenation facilitates the preservation of a more diverse set of representations by directly combining all intermediate features, potentially enhancing the model's capacity to capture complex patterns. On the other hand, feature summation offers a more computationally efficient alternative, reducing the dimensionality of the feature space whilst maintaining essential information through additive residual connections,

[1]With the exception of the third MVM block, which maintains the spatial size, as it has no pooling operation.

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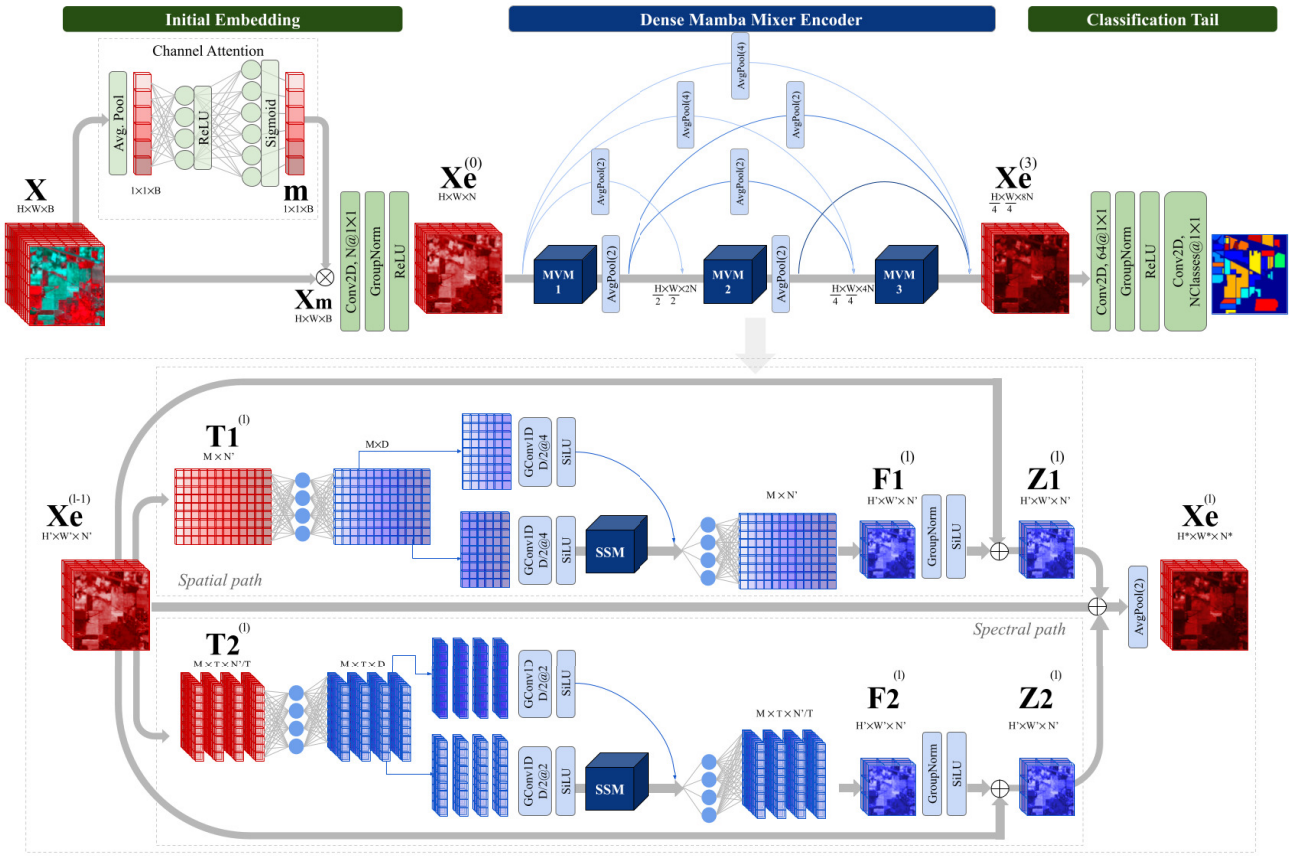


Fig. 3. Graphical depiction of the proposed `DenseMixerMamba` architecture. Regarding the dense mamba mixer encoder, two versions of the dense connection have been developed: i) feature concatenation and ii) residual feature summation.

thus promoting stability and mitigating the risk of overfitting. In this regard, spatial dimensions are $H' = H/2^{l-1}$, $W' = W/2^{l-1}$ and consequently $H^* = H/2^l$, $W^* = W/2^l$, whilst the depth is obtained as $N' = 2^{l-1}N$ and $N^* = 2^l N$, when using feature concatenation, and $N' = N^* = N$ when using feature summation, with $l = \{1, 2, 3\}$. To combine the feature maps, pooling operations with kernels 2×2 and 4×4 are implemented within the dense connections.

Furthermore, each MVM block integrates both spectral and spatial Mamba paths to effectively capture and model intrinsic patterns at both levels. Regarding the spatial paths, input features $\mathbf{Xe}^{(l-1)}$ are flattened into $\mathbf{T1}^{(l)} \in \mathbb{R}^{M \times N'}$ token representation ($M = H'W'$), where each pixel is processed as a token. Tokens are then introduced in the mixer component, and linearly projected into the D-dimensional latent space through an FC layer. Size $D = N'E$ is computed based on the input dimension N' and an expansion factor E. The resulting token representation is split along the state dimension into two parts of size $M \times D/2$ to introduce the token mixer mechanism. In this process, the first part is convolved by a 1-D grouped convolutional kernel (GConv1D) of size $D/2@4$, followed by a SiLU activation function, before being processed by the SSM module to produce $\mathbf{F1}^{(l)}_{\text{SSM}}$. Simultaneously, the second part undergoes a symmetric path without SSM, i.e., only with identical GConv1D and SiLU layers, yielding $\mathbf{F1}^{(l)}_{\text{MIX}}$. It is noteworthy that both GConv1D layers convolve each input channel independently of the others. This is equivalent to performing $D/2$ independent convolutions, each applied on neighbor pixels within a specific channel, thus processing spatial information. Latent features $\mathbf{F1}^{(l)}_{\text{SSM}}$ and $\mathbf{F1}^{(l)}_{\text{MIX}}$ are concatenated to improve the ability to capture the global context, ensuring also that the final feature representation incorporates both sequential and spatial information. The resulting tensor is linearly projected again into the original N'-dimensional feature space. Finally, spatial reshaping is conducted to obtain the feature representation $\mathbf{F1}^{(l)} \in \mathbb{R}^{H' \times W' \times N'}$. To strengthen the learning procedure and mitigate content loss resulting from the input vectorization performed by the MVM block, a residual connection is introduced, which combines the original $\mathbf{Xe}^{(l-1)}$ and feature data $\mathbf{F1}^{(l)}$ into $\mathbf{Z1}^{(l)}$

$$
\begin{aligned}
\mathbf{F1}^{(l)}_{\text{SSM}} &= \text{SSM}\left(\text{SiLU}\left(\text{GConv}\,1D_{\frac{D}{2}}@4\left(\text{FC}_D(\mathbf{T1}^{(l)})_{[:,:D/2]}\right)\right)\right) \\
\mathbf{F1}^{(l)}_{\text{MIX}} &= \text{SiLU}\left(\text{GConv}\,1\,\text{D}_{\frac{D}{2}}@4\left(\text{FC}_D\left(\mathbf{T1}^{(l)}\right)_{[:,D/2:]}\right)\right) \\
\mathbf{F1}^{(l)} &= \text{Reshape}\left[\text{FC}_{N'}\left(\mathbf{F1}^{(l)}_{\text{SSM}} \frown \mathbf{F1}^{(l)}_{\text{MIX}}\right)\right] \\
\mathbf{Z1}^{(l)} &= \mathbf{Xe}^{(l-1)} + \text{SiLU}\left(\text{GroupNorm}\left(\mathbf{F1}^{(l)}\right)\right).
\end{aligned} \tag{7}
$$

A similar procedure is implemented by the spectral branch. In this sense, input $\mathbf{Xe}^{(l-1)}$ is reshaped and flattened into $\mathbf{T2}^{(l)} \in \mathbb{R}^{M \times T \times N'/T}$ token representation ($M = H'W'$), where T is the number of tokens per pixel, i.e., the spectral information of each pixel is rearranged into T tokens of size N'/T. Once more, $\mathbf{T2}^{(l)}$ is linearly projected into the D-dimensional latent state space, with $D = E \cdot N'/T$. Also, the token representation is split along D, resulting in two data volumes of size

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$M \times T \times D/2$. The dual-path is implemented, where the first part is convolved by a 1-D GConv1D of size $D/2@2$ and an SiLU function, before entering the SSM module, and the second part is processed by identical convolution and SiLU layers. In this regard, it must be noted that the convolution operates on pairs of adjacent tokens within a specific channel, thus, facilitating a fine-grained analysis throughout the spectrum of a pixel. Obtained $\mathbf{F2}^{(l)}_{\text{SSM}}$ and $\mathbf{F2}^{(l)}_{\text{MIX}}$, are concatenated and processed in a similar way to obtain the final representation $\mathbf{F2}^{(l)}$. Also, a residual connection is introduced, producing the output feature maps $\mathbf{Z2}^{(l)}$ of the following equation:

$$\begin{aligned}
\mathbf{F2}^{(l)}_{\text{SSM}} &= \text{SSM}\left(\text{SiLU}\left(\text{GConv1D}_{\frac{D}{2}}@4\left(\text{FC}_D\left(\mathbf{T2}^{(l)}\right)_{[:,:D/2]}\right)\right)\right)\\
\mathbf{F2}^{(l)}_{\text{MIX}} &= \text{SiLU}\left(\text{GConv1D}_{\frac{D}{2}}@4\left(\text{FC}_D\left(\mathbf{T2}^{(l)}\right)_{[:,D/2:]}\right)\right)\\
\mathbf{F2}^{(l)} &= \text{Reshape}\left[\text{FC}_{N'}\left(\mathbf{F2}^{(l)}_{\text{SSM}} \frown \mathbf{F2}^{(l)}_{\text{MIX}}\right)\right]\\
\mathbf{Z2}^{(l)} &= \mathbf{Xe}^{(l-1)} + \text{SiLU}\left(\text{GroupNorm}\left(\mathbf{F2}^{(l)}\right)\right) \qquad (8)
\end{aligned}$$

The dual-branch structure inside each path has several beneficial implications. The branch with SSM specializes in integrating complex, sequential features, while the branch without SSM focuses on more independent features. This allows the model to learn complementary data representations. Furthermore, the separation into two branches can also act as a regularization, where each branch learns different aspects of the data. In particular, the "non-SSM" branch can act as a control path, reducing sensitivity to the noise introduced by the SSM block. Finally, by concatenating the outputs of both branches, a more robust propagation of features through the network is facilitated, ensuring that both simple and complex relationships contribute to the overall learning.

The output $\mathbf{Xe}^{(l)}\mathbb{R}^{H^* \times W^* \times N^*}$ is finally obtained by

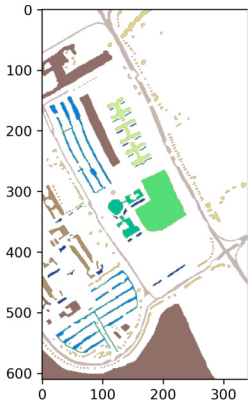
$$\mathbf{Xe}^{(l)} = \text{AvgPool}\left(\mathbf{Xe}^{(l-1)} + \mathbf{Z1}^{(l)} + \mathbf{Z2}^{(l)}\right). \qquad (9)$$

C. Classification Tail

The output of the dense MVM encoder $\mathbf{Xe}^{(3)}$ is refined by a 2-D convolution kernel (Conv2D) with size $64@1 \times 1$ followed by group normalization and ReLU layers. This adjusts the depth of the features, combining the channel information without changing the spatial resolution of the data. Finally, a $N_{\text{classes}}@1 \times 1$ Conv2D adjusts the depth of the data to the number of classes, thus each hyperspectral pixel stores the scores or logits for each class.

D. Network Settings

Regarding network configuration, the parameters H, W, and B correspond to the original dimensions of the input image. The hyperparameter N is initialized to 128 and remains constant throughout the encoder when employing the "feature-summation" version. In contrast, for the "feature-concatenation" version, the depth dimension is progressively increased by a factor of $2^l N$. Extensive experimentation with the expansion factor E has been conducted, leading to a final configuration of $E = 2$ for the spatial branch and $E = 4$ for the spectral branch. Furthermore, in the spectral branch, the number of tokens T has been set to 4 to optimize the representation of the spectral information. Furthermore, Adam optimizer is used with a learning rate $1e^{-3}$, 200 epochs.



Color	Land cover type	Samples
	Background	164624
	Asphalt	6631
	Meadows	18649
	Gravel	2099
	Trees	3064
	Painted metal sheets	1345
	Bare Soil	5029
	Bitumen	1330
	Self Blocking Bricks	3682
	Shadows	947
	Total samples	207400

Fig. 4. Ground truth and samples description of the University of Pavia image.

To address class imbalance during training, the cyclical focal loss (CFL) is adopted as proposed in [64] and [65]. Unlike standard cross-entropy (CE) or fixed-parameter focal loss, CFL dynamically modulates the focusing parameter γ throughout training, alternating between focusing on easy and hard examples. This cyclical adjustment helps prevent early overfitting to dominant classes and has been shown to improve generalization and robustness in classification tasks with imbalanced or noisy data. Its effectiveness and superiority over standard focal loss and CE have already been extensively validated (both theoretically and experimentally) in the hyperspectral domain by works as [66]. In this sense, given the natural imbalance present in HSI datasets, CFL complements the dense sequential modeling capabilities of the Mamba-based architecture by enhancing minority class sensitivity without sacrificing global accuracy.

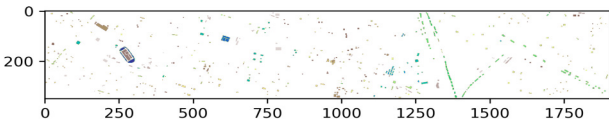
IV. Experimental Discussion

A. Datasets and Experimental Settings

A comprehensive study has been conducted to evaluate the classification performance of the proposed `DenseMixerMamba`. In particular, four different and popular HSI scenes have been used as follows.

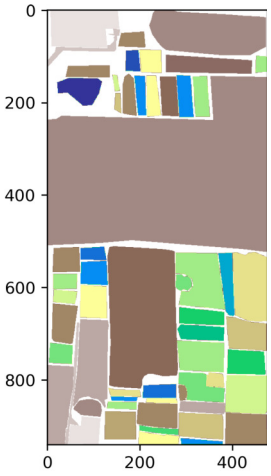
1) The aerial **University of Pavia** scene is a popular classification dataset. It was captured by ROSIS-03 in 2001. It contains 103 spectral bands that cover the spectral range from 430 to 860 nm with a spatial resolution of 1.3 m, and 610×340 pixels. The scene includes nine categories, each corresponding to a different surface material, as shown in Fig. 4. The University of Pavia is characterized by similar spectral properties between some of the features, which makes the classification task more challenging, especially when the spectra between categories are small.
2) The aerial **University of Houston** (UH) scene was acquired in the Houston metropolitan area, Texas, USA. The dataset contains 144 spectral bands that cover the range from 380 to 1050 nm, with a spatial resolution of 2.5 m and 349×1905 pixels. It contains 15 classes of feature types and provides ground-truth labeled data

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Color	Class	Samples	Color	Class	Samples
	Background	649816		Healthy grass	1251
	Stressed grass	1254		Synthetic grass	697
	Tree	1244		Soil	1242
	Water	325		Residential	1268
	Commercial	1244		Road	1252
	Highway	1227		Railway	1235
	Parking lot 1	1233		Parking lot 2	469
	Tennis court	428		Running track	660
	Total samples				664845

Fig. 5. Ground truth and samples description of UH image.

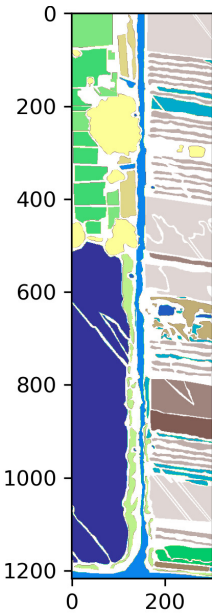


Color	Land cover type	Samples
	Background	59807
	Red roof	14041
	Road	3512
	Bare soil	21821
	Cotton	163285
	Cotton firewood	6218
	Rape	44557
	Chinese cabbage	24103
	Pakchoi	4054
	Cabbage	10819
	Tuber mustard	12394
	Brassica parachinensis	11015
	Brassica chinensis	8954
	Small brassica chinensis	22507
	Lactuca sativa	7356
	Celtuce	1002
	Film covered lettuce	7262
	Romaine lettuce	3010
	Carrot	3217
	White radish	8712
	Garlic sprout	3486
	Broad bean	1328
	Tree	4040
	Total samples	446500

Fig. 6. Ground truth and samples description of the University of HongHu image.

for supervised learning, as shown in Fig. 5. The dataset covers a variety of complex urban feature structures, such as roads, buildings, and vegetation, making the HSI classification task more relevant, but also more difficult.

3) The **HongHu** scene collects spectral signatures of cultivated land located in Honghu City, Hubei Province, China. The data contains 940 × 475 pixels, with 270 bands, and a spectral range of 400–1000 nm. Fig. 6 shows the ground-truth information. The richness of cropland types poses a challenge in HSI classification and validates the proposed classification algorithm for fine agriculture applications.
4) The **HanChuan** dataset collects HSI scenes from the HanChuan city area in HuBei province, China, with 1217 × 303 pixels, 274 bands, and a spectral range of 400–1000 nm. The dataset is located in the rural-urban fringe area and contains a variety of feature types, including cultivated land with seven crop types, water, buildings, and roads. The feature distribution is shown in Fig. 7. The dataset provides detailed ground truth data and is ideal for training and evaluating classification algorithms.



Color	Land cover type	Samples
	Background	111221
	Strawberry	44735
	Cowpea	22753
	Soybean	10287
	Sorghum	5353
	Water spinach	1200
	Watermelon	4533
	Greens	5903
	Trees	17978
	Grass	9469
	Red roof	10516
	Gray roof	16911
	Plastic	3679
	Bare soil	9116
	Road	18560
	Brigth object	1136
	Water	75401
	Total samples	368751

Fig. 7. Ground truth and samples description of the University of HanChuan image.

When testing the proposed `DenseMixerMamba` over each HSI dataset, 30 samples per class have been used to train the model, 10 samples per class have been reserved for validation, and the rest of the samples have been used for testing purposes. Furthermore, two different dense implementations are considered, i.e., *feature concatenation*-based model (**.**) and *feature summation*-based model (**+**).

To make an exhaustive comparison with the current state-of-the-art, 14 different methods have been considered. In particular: three *traditional classifiers*, i.e., random forest (RB) [67], multinomial logistic regression (MLR), and support vector machine (SVM) [68]; four deep neural networks, i.e., CNN2D [66], CNN3D [2], pyramidal ResNet (pResNet) [6], and parameter-free attention network (FreeNet) [69]; six *transformer-based networks*, i.e., standard ViT [48], pixel-wise SpectralFormer ($\text{CAF}_{\text{pixel}}$) and patch-wise SpectralFormer ($\text{CAF}_{\text{patch}}$) [48], spectral–spatial feature tokenization transformer (SSFTT) [17], group-aware hierarchical transformer (GAHT) [51], and groupwise separable convolutional ViT (GSCVIT) [52]. Finally, the spatial–spectral Mamba network (MambaHSI) [57] is considered to compare the behavior of the proposed `DenseMixerMamba`. The training settings of these models have remained faithful to the original works.

The classification results of each model have been evaluated quantitatively and qualitatively. Regarding quantitatively measurements, three accuracy metrics, OA, average accuracy (AA), and the Kappa coefficient have been taken into account. The impact of the number of samples per class on OA has also been evaluated to study the behavior of the proposed network to different amounts of training data. In relation to the qualitative result, classification maps have been produced for each classifier.

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TABLE I

CLASSIFICATION ACCURACY OF DIFFERENT METHODS ON UNIVERSITY OF PAVIA DATASET

Class	Classical Models			Deep Neural Networks				Transformers based						Mamba based		
	RF [67]	MLR [66]	SVM [66]	CNN2D [66]	CNN3D [2]	pResNet [6]	FreeNet [69]	ViT [48]	CAF_{pixel} [48]	CAF_{patch} [48]	SSFTT [17]	GAHT [51]	GSCVIT [52]	MambaHSI [57]	DenseMixerMamba (+)	(.)
1	69.26±3.73	69.58±2.41	72.38±1.35	58.45±24.56	75.65±5.69	85.46±4.49	92.81±3.45	58.05±5.82	65.54±3.89	61.51±8.68	83.23±7.48	91.47±3.92	90.25±6.33	94.64±1.75	**96.02**±2.33	95.99±2.13
2	57.05±6.69	68.7±4.62	77.41±5.59	65.19±10.87	67.08±6.45	90.94±4.49	95.54±1.95	49.37±10.25	52.56±13.17	67.2±13.7	90.65±5.39	92.22±5.06	90.19±4.83	96.26±1.74	**98.4**±1.14	98.2±1.66
3	60.63±6.2	74.42±3.26	76.58±5.2	44.94±15.32	72.13±7.31	87.43±3.57	92.72±3.54	49.85±13.8	44.55±16.17	65.62±11.69	89.04±6.32	92.17±2.98	93.29±3.05	93.2±5.38	93.57±5.22	**94.29**±4.27
4	90.73±4.89	92.33±2.22	92.58±2.21	92.69±7.4	96.81±1.75	96.36±2.15	**98.24**±1.75	96.08±1.85	95.09±1.77	93.04±3.01	98.01±2.26	97.28±1.45	95.47±2.91	89.45±3.94	91.16±3.63	90.78±3.51
5	99.0±0.57	98.87±0.48	99.23±0.29	98.99±1.86	99.76±0.18	99.91±0.23	99.93±0.09	99.21±0.27	99.28±0.28	99.82±0.44	99.82±0.42	99.86±0.15	99.85±0.25	99.99±0.02	99.98±0.07	**100.0**±0.0
6	70.99±6.88	72.96±5.73	82.12±4.19	60.22±18.34	78.97±8.0	87.44±7.45	96.78±2.94	64.65±11.07	66.81±12.54	72.51±9.05	95.37±3.42	94.06±5.0	94.24±4.14	98.6±0.92	98.42±1.57	**98.93**±0.68
7	87.48±4.35	90.72±2.38	91.58±2.07	69.3±27.89	86.65±3.63	92.93±2.02	98.99±0.91	92.86±2.69	89.7±3.72	86.63±4.74	93.27±4.52	97.9±1.69	98.06±4.05	96.36±3.07	**99.19**±0.61	98.64±1.56
8	72.92±5.52	76.73±6.72	79.47±5.86	61.85±23.36	78.81±6.99	86.72±4.24	89.57±9.32	65.0±13.35	71.04±6.73	71.78±11.75	81.77±10.99	89.74±8.48	94.61±6.67	94.22±2.98	**96.81**±1.64	96.22±1.99
9	**100.0**±0.0	99.67±0.13	99.79±0.16	98.36±2.54	99.51±0.48	99.85±0.22	100.0±0.0	99.91±0.07	99.9±0.09	99.54±0.86	98.64±2.03	99.57±0.43	99.64±0.35	99.36±0.62	98.71±1.79	99.23±0.81
OA	67.66±2.26	74.25±2.23	79.98±2.44	66.12±7.26	75.48±2.6	90.05±2.46	95.14±1.32	61.15±3.84	64.06±4.8	71.38±6.18	90.27±2.6	93.03±2.03	92.32±2.38	95.66±0.83	**97.22**±0.54	97.15±0.84
AA	78.67±1.11	82.66±1.53	85.68±1.12	72.22±6.28	83.93±1.16	91.89±1.21	96.06±1.14	75.0±1.26	76.05±1.77	79.74±2.8	92.2±1.56	94.92±0.84	95.07±1.36	95.79±1.14	96.92±0.7	**96.92**±0.72
Kappa(x100)	60.03±2.29	67.57±2.62	74.43±2.85	57.7±8.12	69.24±2.91	86.98±3.12	93.6±1.72	52.84±3.66	56.07±4.65	64.19±6.66	87.34±3.25	90.89±2.55	90.0±3.0	95.25±2.0	97.38±1.31	**97.75**±1.46

TABLE II

CLASSIFICATION ACCURACY OF DIFFERENT METHODS ON UH DATASET

Class	Classical Models			Deep Neural Networks				Transformers based						Mamba based		
	RF [67]	MLR [66]	SVM [66]	CNN2D [66]	CNN3D [2]	pResNet [6]	FreeNet [69]	ViT [48]	CAF_{pixel} [48]	CAF_{patch} [48]	SSFTT [17]	GAHT [51]	GSCVIT [52]	MambaHSI [57]	DenseMixerMamba (+)	(.)
1	91.97±3.79	92.59±3.03	94.21±3.41	93.22±3.8	89.73±6.61	93.07±4.76	94.57±6.05	93.86±6.1	89.24±7.97	94.8±4.73	94.08±6.07	93.88±3.89	91.13±5.28	94.99±4.72	94.17±5.54	**95.93**±4.59
2	93.11±3.52	97.16±1.7	96.87±2.18	95.35±2.73	89.54±6.14	96.71±3.62	**97.67**±2.4	73.84±9.16	78.09±10.42	92.74±5.42	96.93±3.66	97.11±4.27	97.24±3.81	95.67±4.89	96.82±3.63	96.01±3.93
3	94.67±2.02	99.85±0.14	99.63±0.46	99.51±0.55	96.56±2.67	98.69±1.38	**99.85**±0.31	94.2±5.09	96.33±3.3	97.14±3.11	99.42±0.78	99.09±0.99	99.0±1.1	99.8±0.3	99.74±0.3	99.73±0.27
4	94.31±2.92	**97.01**±1.57	96.01±2.23	94.8±2.52	94.66±2.54	96.1±2.16	95.23±3.83	91.46±2.48	92.82±3.91	92.4±3.08	94.44±2.36	95.86±2.63	95.35±2.78	96.85±2.14	96.33±1.85	95.8±3.02
5	93.54±2.62	96.81±1.1	97.99±1.09	98.57±1.23	98.56±0.91	99.55±0.35	99.24±0.86	88.82±6.29	92.55±4.78	97.72±2.75	99.28±0.66	**99.87**±0.14	99.57±0.66	99.31±1.06	99.08±2.3	98.56±4.29
6	92.25±3.74	94.14±4.18	94.91±3.4	**98.81**±1.09	87.75±4.78	94.63±3.72	96.91±4.1	84.42±2.11	86.32±4.42	85.93±4.98	98.53±1.08	96.91±1.52	97.65±2.17	97.44±2.14	97.68±2.07	97.96±1.64
7	75.59±4.4	82.67±4.95	82.21±5.68	88.93±4.1	80.66±5.83	89.05±2.96	90.55±4.69	32.32±9.02	48.63±9.89	74.41±4.95	83.26±4.18	91.11±3.58	89.35±4.39	93.02±3.07	**94.59**±2.38	92.79±4.16
8	62.87±2.92	61.2±4.44	69.78±6.31	80.29±5.26	61.82±8.6	79.06±3.75	80.47±6.37	39.02±7.85	35.51±13.24	73.16±4.45	78.46±6.85	81.91±5.41	**82.43**±6.2	77.5±3.58	81.4±6.87	81.09±5.38
9	71.2±5.26	65.66±7.48	73.16±7.03	87.15±2.91	81.77±2.35	83.37±5.2	88.08±6.61	74.33±5.43	75.07±14.69	80.2±4.09	86.32±4.41	88.56±2.55	87.22±3.99	89.01±2.47	**90.98**±2.81	90.15±3.54
10	68.85±4.44	74.84±6.2	84.59±4.65	89.49±4.32	64.5±8.11	93.32±4.62	90.19±7.9	40.58±11.35	24.46±18.62	85.35±5.28	87.15±9.32	92.53±2.66	87.94±9.99	**96.59**±2.12	95.61±2.46	94.22±4.63
11	68.85±4.59	77.84±4.41	79.47±4.31	85.22±6.54	63.97±4.36	87.35±7.86	89.51±7.3	58.61±8.94	56.7±8.65	80.36±6.05	87.21±5.5	89.97±6.35	87.1±8.72	88.81±4.96	**95.67**±2.52	94.37±4.31
12	55.75±6.7	61.44±4.4	74.85±4.67	**93.29**±3.08	75.71±7.62	91.51±2.12	90.25±4.75	31.12±9.8	28.49±12.22	81.38±4.81	91.92±3.88	93.25±4.02	91.63±5.86	90.64±2.99	90.12±3.32	92.35±3.77
13	37.53±4.24	62.89±4.6	57.95±6.29	95.1±2.2	92.87±2.45	**97.04**±2.85	96.06±2.17	22.63±10.2	24.48±11.65	58.04±4.19	96.36±1.83	96.29±1.08	95.69±3.06	96.81±2.19	96.5±2.49	96.85±2.37
14	95.57±3.23	98.17±1.5	98.48±1.51	99.23±0.77	98.35±0.81	99.84±0.26	99.95±0.16	96.73±1.77	96.24±3.02	96.83±3.19	99.33±1.3	**100.0**±0.0	99.82±0.38	100.0±0.0	100.0±0.0	99.97±0.08
15	95.13±2.26	98.23±0.64	98.94±0.52	99.66±0.48	99.39±0.77	99.84±0.23	**100.0**±0.0	94.84±4.18	97.39±1.81	98.84±0.71	99.5±0.65	99.97±0.1	99.82±0.33	95.34±2.18	95.19±3.76	96.15±2.12
OA	78.76±1.23	82.58±0.92	85.96±0.96	91.95±1.34	82.73±1.71	92.14±1.05	92.78±1.31	65.44±1.9	65.58±2.6	85.89±0.53	91.36±1.4	93.44±1.17	92.17±1.87	93.16±1.17	**94.19**±0.78	93.95±1.31
AA	79.41±1.06	84.03±0.91	86.6±0.9	93.24±1.09	85.06±1.63	93.28±0.83	93.9±1.16	67.79±1.29	68.15±1.9	85.95±0.45	92.81±1.2	94.42±0.99	93.39±1.57	94.12±1.12	**94.93**±0.69	94.79±1.01
Kappa(x100)	77.03±1.33	81.17±0.99	84.81±1.04	91.3±1.45	81.34±1.84	91.5±1.13	92.2±1.41	62.74±2.02	62.93±2.75	84.74±0.57	90.66±1.52	92.91±1.26	91.53±2.02	93.36±2.1	**93.71**±1.94	93.64±1.79

TABLE III

CLASSIFICATION ACCURACY OF DIFFERENT METHODS ON HANCHUAN DATASET

Class	Classical Models			Deep Neural Networks				Transformers based						Mamba based		
	RF [67]	MLR [66]	SVM [66]	CNN2D [66]	CNN3D [2]	pResNet [6]	FreeNet [69]	ViT [48]	CAF_{pixel} [48]	CAF_{patch} [48]	SSFTT [17]	GAHT [51]	GSCVIT [52]	MambaHSI [57]	DenseMixerMamba (+)	(.)
1	62.11±5.58	52.59±4.38	68.18±3.78	82.46±4.18	68.75±2.24	86.94±4.84	77.66±13.03	48.33±3.47	62.04±11.08	80.92±5.06	77.62±5.6	89.21±2.57	86.22±6.61	90.35±5.11	94.94±2.35	**95.44**±1.81
2	31.87±4.94	39.61±3.41	42.4±4.24	71.36±4.39	67.31±3.8	79.24±3.98	67.82±12.39	21.52±9.65	30.95±8.16	62.81±6.35	74.91±5.07	80.54±3.12	78.37±7.89	81.58±5.02	**86.25**±4.9	85.26±3.88
3	60.35±7.18	54.44±5.99	64.75±3.79	69.55±22.45	65.97±6.48	86.06±6.29	82.72±12.45	35.71±15.02	57.92±14.53	71.3±7.22	78.54±6.94	88.85±4.85	88.53±6.86	94.15±4.02	**98.23**±1.16	94.65±2.71
4	82.27±6.8	84.33±3.32	85.91±3.31	89.49±4.97	92.57±1.44	95.74±1.9	97.07±0.85	78.96±12.33	91.81±2.94	92.05±3.25	93.06±2.98	97.42±1.24	96.7±3.71	98.5±1.53	**98.81**±0.94	97.91±1.9
5	71.09±9.8	43.03±4.92	59.3±4.76	97.46±1.23	94.96±2.16	99.13±1.21	96.76±3.26	73.41±6.11	70.56±10.16	91.1±3.17	96.2±2.05	98.22±1.55	98.08±1.36	99.51±0.83	99.8±0.23	**99.95**±0.11
6	23.25±5.05	38.19±4.96	35.98±4.11	54.53±6.58	30.91±6.29	59.41±7.64	72.88±9.88	11.71±4.36	28.86±7.35	43.0±5.24	61.11±9.79	75.78±6.82	70.81±9.74	78.05±7.52	**88.73**±3.04	88.66±3.49
7	79.14±4.0	72.89±4.98	74.88±5.36	88.7±3.33	93.42±1.16	87.4±4.37	87.8±7.0	83.75±5.98	82.72±6.25	91.04±3.28	87.98±5.09	91.59±6.13	87.81±8.15	91.94±5.13	98.74±1.2	**98.78**±1.25
8	46.54±6.22	42.56±5.46	37.45±5.92	55.44±11.26	57.34±2.87	62.96±6.06	71.53±8.15	50.92±6.79	43.78±8.37	49.71±8.51	66.9±4.74	68.31±7.09	68.96±8.7	72.06±6.78	85.61±4.56	**88.84**±5.3
9	26.45±2.52	39.19±4.1	36.29±6.21	53.06±13.09	28.95±4.88	75.23±2.71	79.55±7.72	14.51±8.04	40.66±15.64	60.64±6.19	67.91±12.49	81.2±3.45	81.65±4.76	**88.91**±3.64	88.22±4.92	88.48±5.15
10	70.24±4.93	80.33±3.54	78.73±4.39	84.24±16.16	80.87±2.71	95.7±1.73	93.74±4.83	57.69±11.22	81.96±6.52	90.46±3.84	88.46±5.52	**96.5**±1.3	95.43±3.68	94.51±1.96	92.73±5.09	91.12±4.55
11	87.63±3.9	81.38±3.23	81.17±3.9	87.49±5.6	87.12±6.93	84.86±7.22	80.03±16.25	81.16±8.8	88.4±5.77	90.66±2.28	84.77±8.67	90.47±8.11	91.22±5.2	89.09±6.45	91.83±3.61	**92.05**±2.85
12	51.12±5.55	38.74±6.02	44.98±5.93	79.69±9.47	50.4±12.12	88.18±5.39	81.17±9.02	37.32±10.06	53.89±9.2	78.25±6.63	76.28±11.16	87.72±4.65	82.83±9.13	89.74±6.8	**96.61**±3.41	96.07±2.31
13	43.81±4.19	37.4±5.26	36.17±6.41	57.01±14.26	43.6±4.6	72.54±5.98	65.0±7.66	23.51±4.34	40.11±5.59	57.96±4.72	67.02±7.07	73.33±7.37	72.61±8.58	78.41±4.01	**85.24**±5.16	84.66±4.39
14	56.63±6.63	52.37±7.36	55.58±7.89	68.86±4.76	68.4±7.04	85.31±4.92	71.84±6.77	49.42±7.43	60.86±5.87	73.22±6.26	75.66±4.31	82.74±4.71	78.04±7.43	89.31±3.27	88.88±3.91	**89.36**±3.13
15	71.73±5.42	71.15±3.85	72.58±6.12	94.67±3.59	90.26±2.67	94.78±3.6	95.11±2.41	71.9±5.38	77.17±3.98	86.7±3.68	96.34±1.27	**96.88**±1.63	95.12±0.97	94.2±2.51	95.85±2.12	94.51±3.56
16	81.78±7.84	84.12±3.32	84.99±4.62	91.93±7.75	76.22±5.32	95.64±2.21	89.89±8.15	77.93±7.87	89.99±6.57	97.24±1.46	90.35±6.73	98.27±0.89	93.64±7.78	98.35±1.25	98.35±1.35	**98.83**±0.68
OA	63.76±2.37	62.75±1.54	66.23±2.19	79.18±6.25	69.57±1.85	86.08±1.91	80.93±5.68	55.68±2.96	67.44±3.23	80.29±2.09	80.85±2.64	88.73±1.02	86.08±1.94	90.37±1.45	93.44±0.95	**93.6**±0.99
AA	59.13±0.87	57.02±1.26	59.96±1.48	76.62±6.22	68.56±1.1	84.32±2.23	81.91±3.84	51.11±2.35	62.61±2.69	76.07±2.08	80.19±2.32	87.31±1.02	85.38±1.17	89.29±1.42	**93.05**±0.74	92.79±0.96
Kappa(x100)	58.77±2.43	57.56±1.65	61.34±2.38	75.99±6.95	65.39±1.95	83.84±2.19	78.08±6.27	49.73±3.19	62.75±3.5	77.16±2.38	77.91±2.91	86.87±1.18	83.87±2.15	89.07±2.48	**93.13**±2.09	91.8±2.94

TABLE IV

CLASSIFICATION ACCURACY OF DIFFERENT METHODS ON HONGHU DATASET

Class	Classical Models			Deep Neural Networks				Transformers based						Mamba based		
	RF [67]	MLR [66]	SVM [66]	CNN2D [66]	CNN3D [2]	pResNet [6]	FreeNet [69]	ViT [48]	CAF_{pixel} [48]	CAF_{patch} [48]	SSFTT [17]	GAHT [51]	GSCVIT [52]	MambaHSI [57]	DenseMixerMamba (+)	(.)
1	71.9±3.85	75.28±3.12	81.45±3.89	67.18±10.67	80.07±4.37	94.23±1.9	90.32±10.48	82.61±6.23	84.2±3.99	90.71±3.79	88.93±6.08	93.74±2.26	90.99±4.26	96.59±1.45	97.03±1.18	**97.48**±0.57
2	74.9±2.92	72.93±3.51	79.18±3.86	50.28±28.47	86.35±4.78	92.32±3.97	92.24±4.75	73.82±6.72	78.2±4.42	84.8±4.61	89.94±4.13	90.3±5.66	91.73±3.77	95.53±2.4	96.37±3.16	**96.95**±1.73
3	70.66±4.13	76.74±3.15	71.44±3.08	38.72±40.42	78.93±2.93	83.84±3.7	84.33±5.38	71.92±4.97	70.67±4.91	78.69±4.07	83.32±4.69	83.48±5.8	78.06±4.5	94.62±1.18	**94.96**±1.01	94.71±1.27
4	63.8±3.72	67.04±4.95	70.05±4.18	78.02±15.62	83.07±3.37	94.9±1.01	89.44±4.51	56.17±6.39	64.57±7.58	91.11±2.72	88.31±4.45	95.36±1.75	93.12±2.8	**97.65**±1.01	97.34±1.32	97.41±0.89
5	58.85±4.13	62.48±4.02	65.85±2.84	46.23±29.09	76.11±5.9	90.39±6.18	76.52±26.55	62.42±5.11	68.87±3.78	84.45±2.15	85.61±4.78	90.32±5.81	89.81±8.13	96.43±1.43	96.89±1.84	**97.24**±1.86
6	69.75±3.22	80.59±2.03	70.95±4.81	65.19±30.85	88.46±1.53	90.72±2.9	82.35±26.25	71.28±7.22	76.92±3.22	83.49±7.78	89.36±3.58	92.48±2.72	91.22±4.46	**94.39**±2.47	93.98±2.23	94.21±2.28
7	38.22±5.92	50.07±3.42	49.35±3.59	24.72±22.23	58.55±5.44	71.1±4.97	61.16±22.45	48.7±5.94	46.71±9.52	55.18±6.7	65.18±8.03	76.57±4.65	74.45±6.96	87.15±3.15	**88.01**±3.11	88.01±2.82
8	16.41±5.45	25.93±3.1	25.35±3.43	20.47±14.09	39.43±5.71	60.38±5.88	68.65±23.53	25.73±5.59	26.95±7.37	37.99±6.76	69.87±8.78	71.05±9.05	70.24±8.24	96.08±3.28	96.59±1.74	**97.39**±2.92
9	84.14±2.44	88.87±1.2	85.82±2.4	63.23±38.75	92.58±1.57	96.03±1.86	86.08±28.75	90.13±3.68	91.08±1.33	91.37±3.83	94.27±2.27	**96.52**±1.45	96.18±1.34	94.88±1.74	96.11±1.37	96.39±1.56
10	23.83±5.16	37.18±4.66	37.46±2.85	28.05±28.45	59.33±7.19	81.06±4.46	76.72±26.03	21.88±6.28	35.97±8.55	58.88±8.58	81.08±4.35	85.27±4.66	82.04±6.04	90.72±2.91	**93.27**±2.1	93.07±2.14
11	24.53±5.74	26.47±2.83	32.58±2.74	18.0±13.24	53.94±4.31	71.41±6.95	71.43±23.93	24.74±6.25	34.29±7.12	47.46±11.88	73.26±4.78	82.2±4.7	80.03±5.2	**92.75**±3.04	91.27±3.08	89.38±3.28
12	55.85±4.48	33.83±10.79	54.82±3.4	4.62±8.32	62.46±6.11	75.14±3.36	70.53±24.21	55.63±4.41	51.72±6.44	67.8±4.3	70.2±6.9	73.62±7.11	74.34±8.19	87.26±4.95	88.13±5.29	**88.95**±5.22
13	48.19±6.95	34.53±3.28	44.37±5.22	36.02±17.36	60.98±4.71	71.03±2.46	68.16±23.45	41.42±6.88	47.71±6.39	60.1±6.83	71.03±6.34	76.78±3.88	72.39±3.13	**88.91**±3.0	88.52±3.25	88.27±2.68
14	50.59±2.05	41.01±10.95	56.02±2.56	26.13±25.43	78.38±4.89	87.77±6.63	79.14±25.04	62.29±2.34	62.43±5.68	71.1±6.11	89.72±3.85	93.75±2.15	91.08±3.98	94.02±1.8	**95.0**±1.46	94.85±1.48
15	72.14±2.53	80.52±2.68	78.42±1.26	49.99±32.4	89.74±3.83	96.73±1.27	91.55±11.3	74.42±7.75	79.69±3.36	91.69±3.3	95.53±1.44	98.01±1.19	97.69±1.11	98.92±1.15	99.0±1.18	**99.18**±0.78
16	75.19±4.69	66.51±4.38	73.75±3.74	47.17±41.77	81.07±7.04	91.84±4.2	85.66±28.61	64.16±8.49	72.28±6.54	79.64±5.71	92.02±4.52	92.25±5.49	92.98±4.29	96.94±1.64	**98.12**±0.95	98.02±1.41
17	67.15±3.59	59.78±6.09	72.2±4.6	50.74±37.4	78.79±5.55	93.08±4.25	91.89±6.43	63.8±10.1	69.86±8.82	88.2±2.73	89.49±7.58	95.88±2.95	93.2±3.05	98.45±1.78	99.25±1.2	**99.37**±1.27
18	63.81±3.24	67.36±3.25	60.74±4.04	48.96±23.17	78.97±6.38	94.49±2.2	88.18±28.29	58.78±9.19	53.26±12.95	77.75±4.78	91.15±6.04	95.52±3.05	94.52±4.03	96.58±1.82	**98.35**±0.86	98.31±0.98
19	62.5±3.62	71.08±3.12	59.35±4.46	19.67±28.98	83.16±3.21	92.71±2.03	90.97±2.74	64.38±6.15	74.86±5.54	79.43±10.77	85.32±5.66	91.92±1.94	91.05±3.1	91.32±2.74	93.04±1.75	**93.27**±1.66
20	73.22±2.63	70.49±2.76	71.21±2.61	53.17±26.82	89.62±2.46	95.58±1.4	81.41±27.38	68.32±5.67	72.87±3.85	88.69±5.57	91.38±3.77	94.66±1.98	95.32±3.09	99.11±0.4	99.16±0.72	**99.54**±0.37
21	67.42±6.56	49.98±5.01	66.1±4.43	48.39±41.68	76.26±7.99	93.99±6.32	90.45±25.16	51.94±11.74	51.31±7.82	85.45±5.86	87.72±9.16	91.23±7.73	93.49±4.84	99.93±0.15	**100.0**±0.0	100.0±0.0
22	69.11±4.21	71.08±3.7	68.37±4.42	55.18±33.85	88.14±2.38	95.38±1.5	92.23±4.43	66.15±9.38	65.96±7.29	87.21±6.91	90.19±5.56	95.84±2.86	92.83±7.8	99.5±0.53	**99.67**±0.46	99.3±0.9
OA	60.31±1.96	63.37±2.07	64.89±1.92	57.43±9.14	78.11±1.82	88.66±0.87	83.26±11.63	58.1±3.04	63.79±2.92	80.88±2.36	84.71±2.43	90.33±0.95	88.17±1.5	94.98±0.51	95.08±0.64	**95.12**±0.55
AA	59.19±0.66	59.53±0.86	62.49±0.6	42.73±5.8	75.65±1.67	87.01±1.12	82.25±17.52	59.12±1.57	62.74±0.91	76.42±2.91	84.68±1.52	88.94±0.73	87.58±0.66	94.9±0.5	95.46±0.43	**95.51**±0.28
Kappa(x100)	53.61±1.9	56.85±2.0	58.38±1.92	49.27±9.03	73.38±2.04	85.84±1.07	79.3±14.44	51.78±2.88	57.55±2.8	76.36±2.84	81.21±2.82	87.91±1.14	85.3±1.76	93.62±2.32	**95.05**±1.42	94.67±1.24

B. Experimental Results

1) Quantitative Evaluation of Classification Performance: In this section, classification experiments were conducted on four popular datasets: University of Pavia, UH, HanChuan, and HongHu, to compare the proposed `DenseMixerMamba` model with existing methods. The classification results were quantitatively analyzed using three accuracy metrics: OA, AA, and the Kappa coefficient. The mean and standard deviation results of 5 Monte Carlo runs are shown in Tables I–IV. The best results in each table are highlighted in bold.

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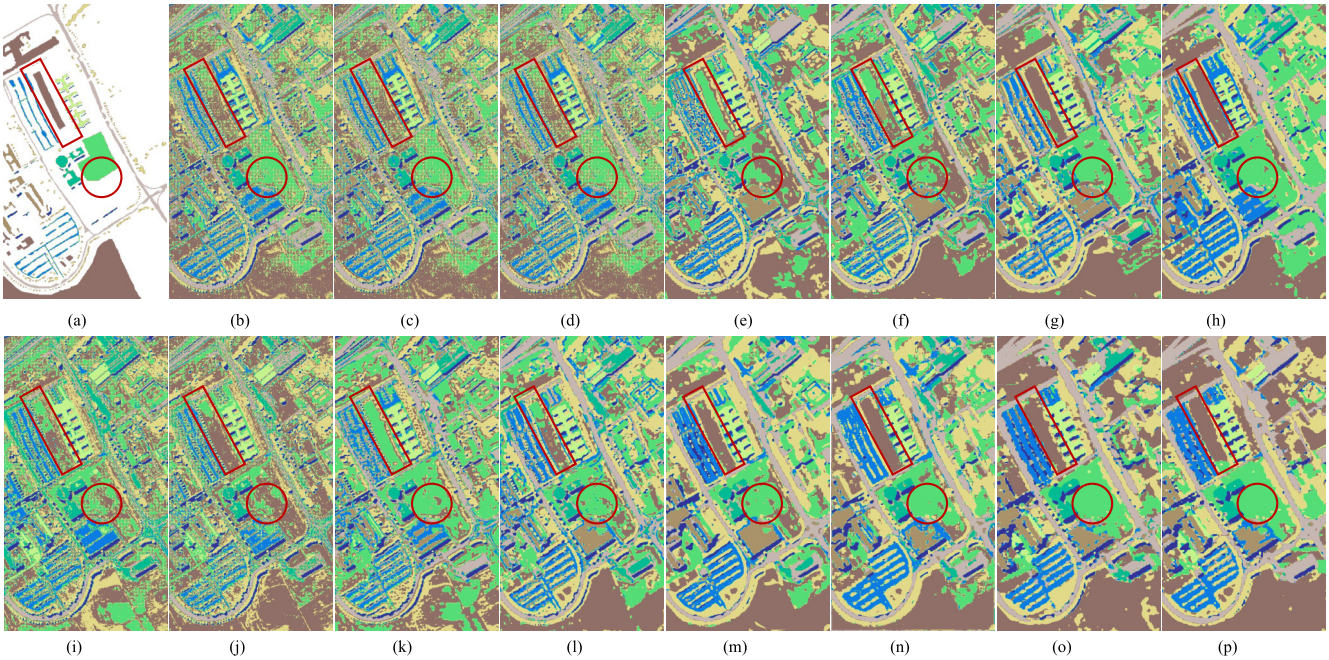


Fig. 8. Classification maps obtained by different methods on the University of Pavia dataset. (a) GT. (b) RF. (c) MLR. (d) SVM. (e) CNN2D. (f) CNN3D. (g) pResNet. (h) FreeNet. (i) ViT. (j) CAF_{pixel}. (k) CAF_{patch}. (l) SSFTT. (m) GAHT. (n) GSCVIT. (o) MambaHSI. (p) DenseMixerMamba.

Generally, the DL-based classifiers present higher accuracy than the traditional methods according to Tables I–IV, indicating the significant superiority of deep features for HSI classification. With regard to deep neural networks and transformer-based networks, the performance of early-developed classifiers is not satisfactory, whereas recent approaches have considerably improved the classification accuracy. Remarkably, the proposed `DenseMixerMamba` model achieves the highest classification accuracy among all the competitors in the four considered datasets. Specifically, the optimal OA of the `DenseMixerMamba` algorithm outperforms the optimal OA of the transformer-based algorithm by 4.19%, 0.75%, 4.87%, and 4.95% in the four datasets, respectively. The smaller deviations of the proposed network also indicate a more stable behavior than the transformer methods. Compared to the Mamba classifiers in the four datasets, the `DenseMixerMamba` improves the OA by approximately one and a half percentage points on average. Overall, the designed `DenseMixerMamba` achieves approximately a classification accuracy of 97%, 94%, 93%, and 95% in the four data, respectively.

Moreover, three mamba-based algorithms, including MambaHSI, `DenseMixerMamba`(+), and `DenseMixerMamba`(.), achieved the highest classification accuracy in most classes, which shows the benefits of Mamba in HSI classification tasks. In addition, the proposed `DenseMixerMamba` algorithm with both feature concatenation and feature summation consistently outperforms the MambaHSI model in all datasets, with an increase in the OA of approximately one percentage point. These results substantiate the contribution of dense connections and the MVM block in the improved `DenseMixerMamba` network.

2) Visual Assessment of Classification Results: The classification maps of the proposed algorithm and the comparison algorithms in the four datasets are shown in Figs. 8–11. The red boxes added to the classification maps highlight the differences in classification in local regions. By analyzing these highlighted areas, it is evident that `DenseMixerMamba` consistently reduces misclassification in all four datasets, particularly in boundary regions and highly mixed areas. This improvement can be attributed to the efficient modeling of long-range dependencies performed by the Mamba structure, which mitigates errors caused by feature ambiguity, and the integration of multiscale information provided by the proposed DenseMixer structure, which enhances class feature differentiation and thereby reduces the probability of misclassification. Taking the University of Pavia dataset as an example, in the bare soil region, traditional methods such as SVM, MLR, and RF exhibit severe class intermixing, particularly in fine-grained areas where misclassified pixels are prevalent. CNN2D and CNN3D reduce noise to some extent, but CNN2D fails to capture spatial information, leading to blocky classification results, while CNN3D struggles with class confusion in certain areas. In the meadow region,CAF_{patch} and CAF_{pixel} produce classification errors within the red-boxed regions, with unclear boundaries. ViT and GSCVIT suffer from misclassification issues in the small area. In contrast, `DenseMixerMamba` significantly reduces misclassification and confusion, as its dual-branch MVM block architecture effectively extracts complementary local and global features, leading to higher classification consistency and robustness. For the Houston dataset, in the red-boxes that highlight the road and railway areas, `DenseMixerMamba` ensures better class consistency, producing classification results that are closer to the ground truth compared to other methods. By leveraging multiscale information integration in the DenseMixer structure, the model enhances feature differentiation, which is particularly beneficial for distinguishing spectrally similar classes. By analyzing

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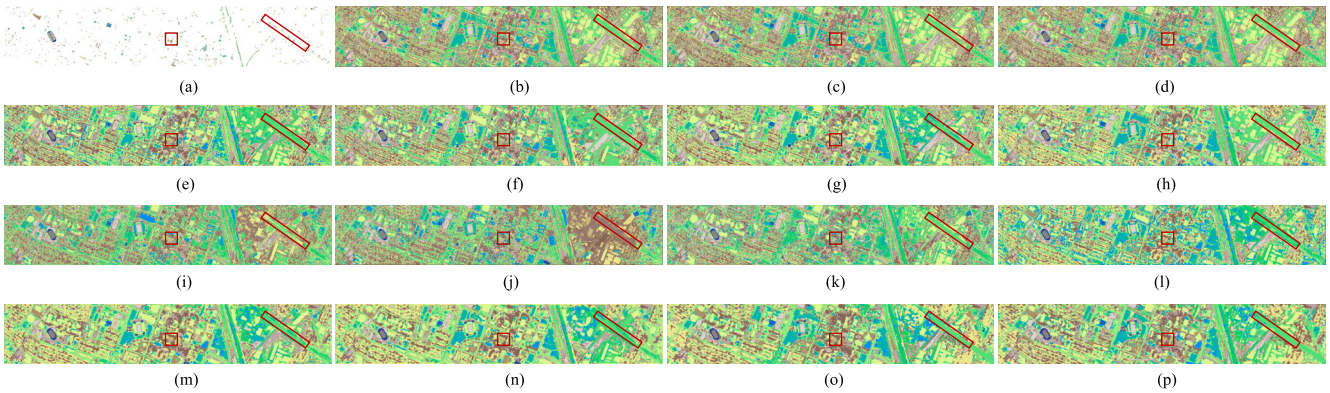


Fig. 9. Classification maps obtained by different methods on the Houston dataset. (a) GT. (b) RF. (c) MLR. (d) SVM. (e) CNN2D. (f) CNN3D. (g) pResNet. (h) FreeNet. (i) ViT. (j) CAF_{pixel}. (k) CAF_{patch}. (l) SSFTT. (m) GAHT. (n) GSCVIT. (o) MambaHSI. (p) DenseMixerMamba.

the distribution of synthetic grass and healthy grass, it is evident that the proposed model exhibits stronger recognition capabilities for small-scale areas.

`DenseMixerMamba` demonstrates superior adaptability in complex regions. In the HanChun dataset, traditional methods suffer from frequent misclassification in the tree-covered areas, and all comparison methods suffer from the problem of local classification mutation, `DenseMixerMamba` mitigates this problem through its densely connected MVM-based encoder, ensuring a smoother classification transition and stronger interclass discrimination. In bare soil, strawberry, and cowpea regions, which feature complex terrain, MambaHSI tends to overemphasize local features, leading to increased misclassification pixels, whereas `DenseMixerMamba` better preserves intricate regional details, yielding more stable classification results. For the HongHu dataset, which contains diverse crop types, in the large cotton planting areas, `DenseMixerMamba` exhibits minimal misclassification, benefiting from its spectral attention mechanism that dynamically selects relevant spectral bands. For small agricultural plots, the classification results clearly show that compared to traditional methods, which produce excessive class smoothing and blurry boundaries, `DenseMixerMamba` ensures more precise classification in boundary regions. Unlike ViT and GAHT, which introduce boundary confusion in complex regions, or MambaHSI, which exhibits over-smoothed boundaries, `DenseMixerMamba` maintains better boundary continuity, producing classification maps with higher interpretability and spatial coherence.

In general, these classification maps in four different datasets demonstrate that `DenseMixerMamba` outperforms comparison algorithms in classification accuracy, boundary clarity, and adaptability to complex regions, validating its effectiveness in land cover classification tasks.

3) Classification Performance of Different Samples: This experiment evaluates the classification accuracy of the algorithms under different sample conditions, with the sample size set at 5, 10, 20, 30, and 40, respectively. These samples were randomly selected, ensuring consistency in all classifiers to guarantee a fair comparison. The results are shown in Fig. 12, it can be observed that in these four datasets, the proposed algorithm achieves an OA above 70% under the small sample condition of 5 samples. When the sample size exceeds 30, the `DenseMixerMamba` algorithm attains the highest classification accuracy compared to other algorithms, with an accuracy exceeding 94% in the four datasets. Analyzing the accuracy curves for all algorithms in these four datasets, it is evident that the proposed algorithm shows smoother changes in classification accuracy with the variation in sample size, indicating the best stability among all tested algorithms.

TABLE V

NUMBER OF TRAINABLE PARAMETERS, IMAGES PROCESSED PER SECOND, AND PIXELS PROCESSED PER SECOND FOR CONSIDERED METHODS

Type	Parameter (K) (↓)	Images per second (↑)	Pixels per second (↑)
Deep Neural Networks			
CNN2D [66]	1485	**1.35**	**279,360.55**
CNN3D [2]	**101**	0.32	65,935.26
pResNet [6]	1110	0.07	14,988.99
FreeNet [69]	221	0.04	8,369.03
Transformers based			
ViT [48]	162	0.11	23,491.36
CAF_{pixel} [48]	162	0.09	17,768.78
CAF_{patch} [48]	184	0.10	20,373.15
SSFTT [17]	489	**0.36**	**73,764.70**
GAHT [51]	927	0.15	29,967.41
GSCViT [52]	**153**	0.17	34,462.35
Mamba based			
MambaHSI [57]	412	4.18	866,932.00
`DenseMixerMamba` (+)	**282**	**5.00**	**1,037,000.00**
`DenseMixerMamba` (.)	1749	3.78	783,972.00

4) Comparative Analysis of Model Complexity and Parameter Efficiency: Table V provides a comprehensive comparison of the selected models used for HSI classification, classified into deep neural networks, transformer-based models, and Mamba-based models, with the number of parameters (in thousands), images processed per second, and pixels processed per second as a measure of computational complexity. These metrics have been selected to make a fair comparison between models that perform patch processing and models that conduct full-image processing, comparing models at the level of processed data.

Among deep neural networks, the CNN2D exhibits the highest number of parameters (1485K) and also achieves the highest processing speed among non-Mamba-based architectures. This indicates that, despite its large parameter count, the convolutional operations are highly optimized, resulting in strong performance in terms of both pixels processed and images per second. However, compared to the lightweight `DenseMixerMamba` (+), the CNN2D model contains 5.27× more parameters and is approximately 3.7× slower in terms of

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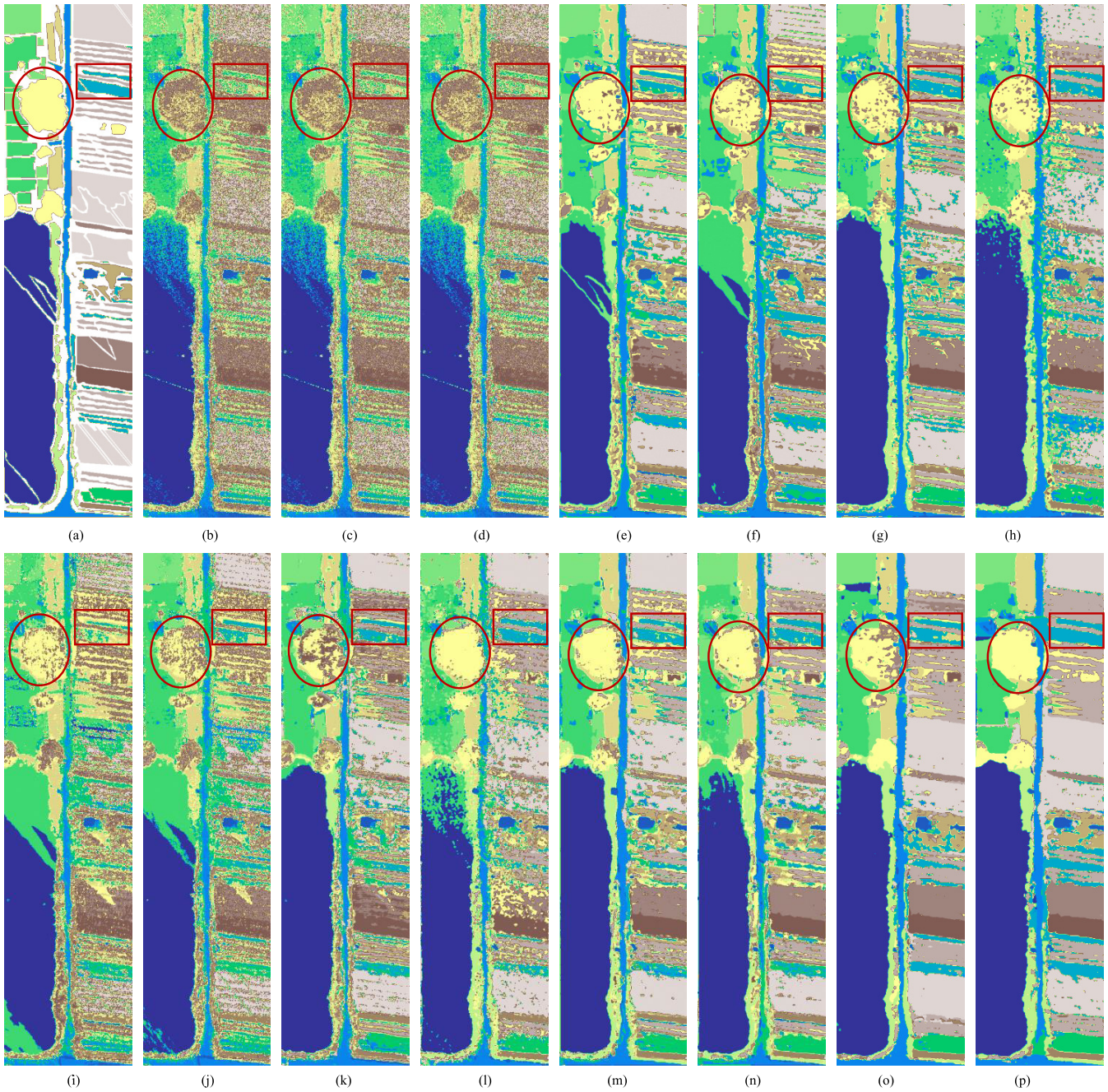


Fig. 10. Classification maps obtained by different methods on the HanChuan dataset. (a) GT. (b) RF. (c) MLR. (d) SVM. (e) CNN2D. (f) CNN3D. (g) pResNet. (h) FreeNet. (i) ViT. (j) CAF_{pixel}. (k) CAF_{patch}. (l) SSFTT. (m) GAHT. (n) GSCVIT. (o) MambaHSI. (p) DenseMixerMamba.

throughput.On the other hand, the CNN3D model is the most parameter-efficient among all evaluated architectures, with only 101K parameters. Nevertheless, this compactness comes at the cost of inference speed: it is approximately 15.6× slower than `DenseMixerMamba` (+) in terms of image throughput, and it processes pixels at a rate 15.7× lower. These results suggest that, although 3-D convolutions offer spectral–spatial modeling capabilities, they introduce substantial computational overhead that limits real-time or high-throughput applications. The pResNet model, with 1110K parameters, is among the heaviest CNN-based architectures analyzed, behind only to the CNN2D. Despite its increased capacity, it achieves the lowest throughput within the CNN family, as it is approximately 71.4× slower than `DenseMixerMamba` (+) in image processing and 69.2× slower in pixel throughput. This suggests that the increased depth and complexity of the residual blocks significantly increased the complexity of the model, negatively affecting performance. Similarly, FreeNet, with a moderate parameter count of 221K, demonstrates even lower performance than the CNN3D, achieving an image processing speed 125× slower than `DenseMixerMamba` (+) and a pixel throughput 128.9× lower. Although compact, the architectural design appears to be suboptimal for efficient inference, highlighting a significant tradeoff between lightweight design and execution speed in convolutional models.

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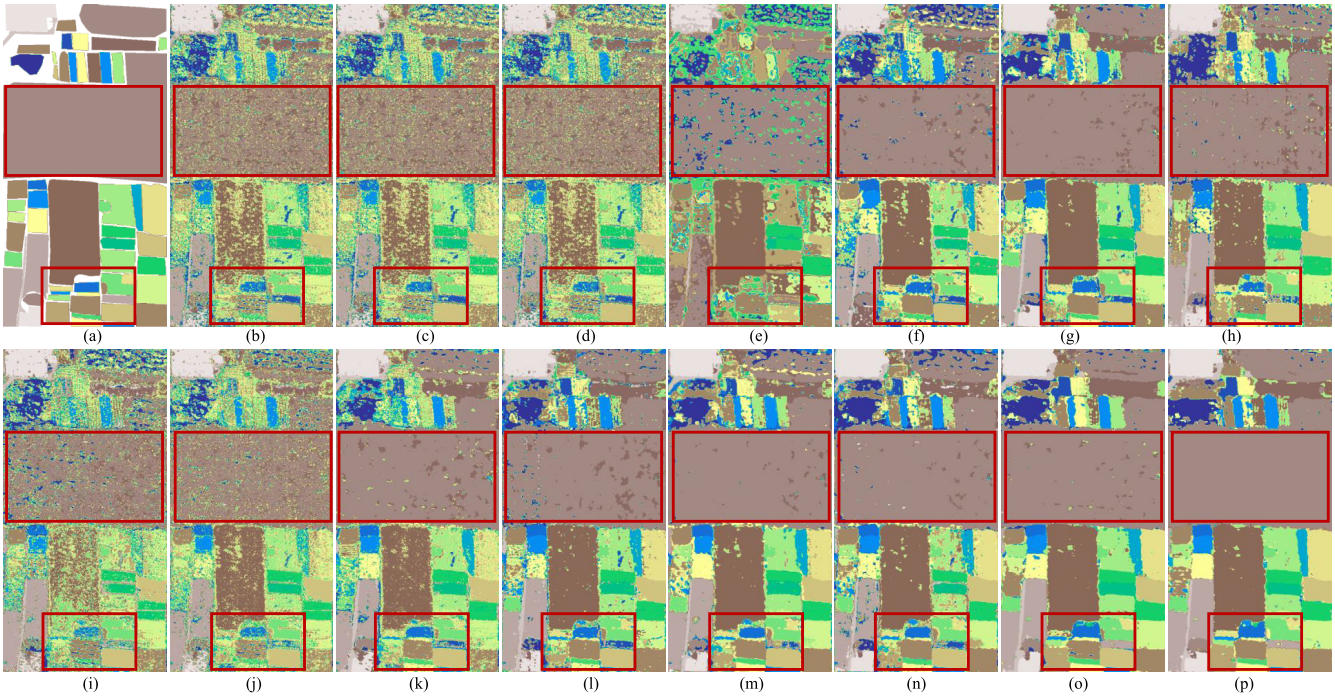


Fig. 11. Classification maps obtained by different methods on the HongHu dataset. (a) GT. (b) RF. (c) MLR. (d) SVM. (e) CNN2D. (f) CNN3D. (g) pResNet. (h) FreeNet. (i) ViT. (j) $\text{CAF}_{\text{pixel}}$. (k) $\text{CAF}_{\text{patch}}$. (l) SSFTT. (m) GAHT. (n) GSCVIT. (o) MambaHSI. (p) DenseMixerMamba.

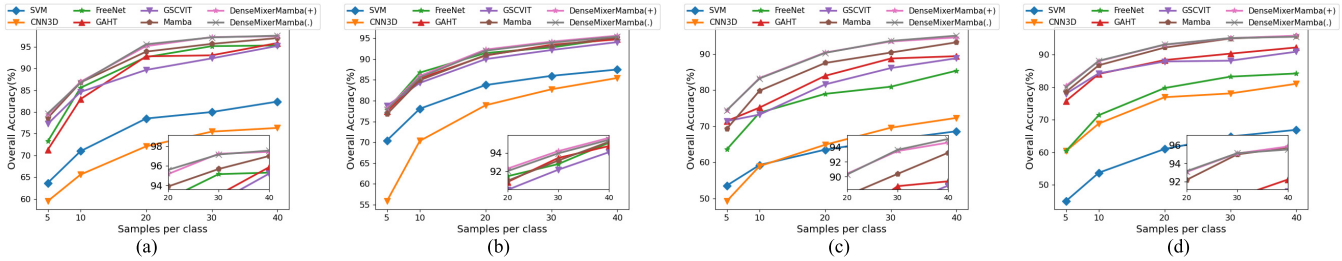


Fig. 12. Evolution of the OA regarding the number of samples per class used during training. (a) University of Pavia. (b) University of Houston. (c) HanChuan. (d) HonghHu.

Regarding transformer-based models, ViT, $\text{CAF}_{\text{pixel}}$, $\text{CAF}_{\text{patch}}$, and GSCViT exhibit relatively low parameter counts, ranging from 153K to 184K. However, this parameter efficiency does not translate into high throughput: ViT, $\text{CAF}_{\text{pixel}}$, $\text{CAF}_{\text{patch}}$ models are between 55.6× and 45.5× slower than `DenseMixerMamba` (+) in image processing, and around 44.1 × −58.4× slower in pixel throughput, whilst GSCViT is 29.4× slower when processing images, reaching a pixel/s rate 30.1× worse than the `DenseMixerMamba` (+). In contrast, the SSFTT model achieves a better tradeoff between complexity and efficiency. With 489K parameters, it attains 0.36 images/s and 73,764.70 pixels/s. In this sense, it exhibits 13.9× and 14.1× lower throughput than `DenseMixerMamba` (+), respectively. This makes SSFTT the most efficient model among the transformer-based architectures evaluated, offering a better compromise between model size and inference speed. In contrast, GAHT has the highest parameter count among transformers (927K) but only moderate computational performance, reaching 0.15 images/s and 29 967.41 pixels/s. As a result, it is approximately 33.3× slower in image processing and 34.6× slower in pixel throughput relative to `DenseMixerMamba` (+). Despite its larger capacity, GAHT does not outperform lighter transformer variants.

Finally, the Mamba-based models exhibit a substantial advantage by combining low parameter counts with exceptional computational efficiency. The base MambaHSI model includes 412K parameters and achieves 4.18 images per second and 866 932 pixels per second, already surpassing all non-Mamba architectures in throughput. The proposed lightweight variant, `DenseMixerMamba` (+), further reduces the parameter count to just 282K while improving inference performance to 5.00 images/s and 1 037 000 pixels/s. Compared to GSCViT, i.e., one of the most efficient transformer-based models, the proposed `DenseMixerMamba` (+) is approximately 29.4× faster in image processing and 30.1× faster in pixel throughput, although it has 1.84× more parameters. On the other hand, the full `DenseMixerMamba` (.) model comprises 1749K parameters but still achieves high throughput with ratios of 3.78 images/s and 783 972 pixels/s, respectively. This makes `DenseMixerMamba` (.) 22.2× faster than `pResNet` and 11.5× faster than `GSCViT` in pixel

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TABLE VI
ABLATION STUDY ON FEATURE INTEGRATION

DenseMixerMamba (+)

	OA(%)	AA(%)	Kappa(x100)
Spectral	80.81 ± 2.69	79.29 ± 1.79	76.40 ± 4.38
Spatial	93.06 ± 0.89	92.76 ± 0.70	93.13 ± 1.33
Spectral+Spatial	93.44 ± 0.95	93.05 ± 0.74	93.13 ± 2.09

DenseMixerMamba (.)

	OA(%)	AA(%)	Kappa(x100)
Spectral	82.33 ± 2.04	80.32 ± 1.45	77.40 ± 3.55
Spatial	92.72 ± 1.50	92.41 ± 1.20	92.40 ± 1.72
Spectral+Spatial	93.60 ± 0.99	92.79 ± 0.96	91.80 ± 2.94

throughput, highlighting its suitability for more complex tasks without compromising efficiency.

These results make Mamba-based architectures, especially `DenseMixerMamba` (+), highly suitable for next-generation HSI analysis, highlighting their role in balancing computational demands and classification accuracy.

C. Ablation Study on Spectral, Spatial, and Combined Feature Integration

To thoroughly evaluate the contributions of key components within the proposed `DenseMixerMamba` network, considering both implementations, i.e., feature summation (+) and feature concatenation (.), an ablation study is conducted to analyze their individual and combined effects on classification performance. Several configurations of the `DenseMixerMamba` (+) and `DenseMixerMamba` (.) are examined in the HanChuan dataset, due to its spectral complexity caused by its numerous shaded areas. These configurations include: i) a spatial-only version that processes spatial features while excluding spectral information; ii) a spectral-only version that focuses solely on spectral features without spatial processing; and iii) a joint spectral–spatial version that integrates both feature types for combined analysis. This study provides a comprehensive understanding of how the integration of spectral and spatial branches contributes to improved classification accuracy, feature extraction, and overall model performance.

Table VI provides the comparison. The results demonstrate that the spatial-only configuration significantly outperforms the spectral-only configuration in terms of OA, AA, and the Kappa coefficient for both architectures. This highlights the dominant role of spatial information in the capture of structural and contextual patterns. However, the combined spectral–spatial configuration consistently achieves the highest performance, with `DenseMixerMamba` (+) achieving an OA of 93.44% and `DenseMixerMamba` (.) achieving 93.60%. These results emphasize that the integration of spectral and spatial features is critical for maximizing classification accuracy by leveraging complementary information from both dimensions. The marginal differences between the two architectures suggest comparable effectiveness in integrating these features, underscoring the overall robustness of the proposed design. Nonetheless, the significant difference in the number of parameters makes the (+) model more appropriate than the (.) version.

TABLE VII
IMPACT OF LOSS ON MODEL ACCURACY

Class	`DenseMixerMamba` (+)		`DenseMixerMamba` (.)	
	CE	CFL	CE	CFL
1	95.68±1.74	**96.02**±2.33	95.61±1.66	**95.99**±2.13
2	97.34±1.3	**98.40**±1.14	96.98±2.13	**98.20**±1.66
3	92.32±4.53	**93.57**±5.22	92.76±4.89	**94.29**±4.27
4	89.56±2.07	**91.16**±3.63	89.69±2.45	**90.78**±3.51
5	99.97±0.07	**99.98**±0.07	99.99±0.02	**100.0**±0.0
6	**99.07**±0.92	98.42±1.57	99.26±0.65	**98.93**±0.68
7	98.64±1.62	**99.19**±0.61	98.44±1.64	**98.64**±1.56
8	96.81±1.5	96.81±1.64	**96.31**±2.08	96.22±1.99
9	98.19±1.67	**98.71**±1.79	97.95±1.81	**99.23**±0.81
OA	96.58±0.72	**97.22**±0.54	96.41±1.06	**97.15**±0.84
AA	96.40±0.74	**96.92**±0.7	96.33±0.8	**96.92**±0.72
Kappa(x100)	94.88±2.27	**97.38**±1.31	94.75±2.29	**97.75**±1.46

D. Impact of Loss on Model Accuracy

To finalize the evaluation of the proposed model, `DenseMixerMamba`(+) and `DenseMixerMamba`(.) have been tested using different loss functions. In particular, CFL and CE have been compared on the University of Pavia dataset. Table VII provides the results; 10 Monte Carlo runs have been performed to obtain the mean and deviation values.

When trained using the standard CE loss (hereinafter, CE-models), both models demonstrate strong and consistent performance across all classes. The lightweight variant, `DenseMixerMamba` (+), slightly outperforms the deeper `DenseMixerMamba` (.) in OA (96.58% versus 96.41%) and AA (96.40% versus 96.33%), despite having significantly fewer parameters. This indicates that the proposed efficient architecture is not only compact but also highly effective. Focusing on a per-class analysis, CE-models achieve particularly high accuracy for stable classes such as class 5-*painted metal sheets* (99.97%–99.99%), class 6-*bare soil* (98.42%–99.07%), and class 7-*bitumen* (98.44%–98.64%), which also have a good number of labeled samples, i.e., 1345, 5029, 1330, respectively. However, in more challenging categories, such as class 3-*gravel* and class 4-*trees*, CE-based performance is comparatively lower, especially in the standard deviations, suggesting limited robustness to intraclass variability.

On the other hand, the use of CFL leads to consistent performance improvements across almost all classes and metrics for both model variants. Compared to CE, CFL enhances the accuracy in difficult classes such as class 3-*gravel* and class 4-*trees*, where improvements of over 1.2%–1.6% are observed. The `DenseMixerMamba` (.) achieves the best class-specific results in categories such as class 3-*gravel* (94.29%), class 5-*painted metal sheets* (100.0%), and class 9-*shadows* (99.23%), and also reports the highest Kappa index (97.75), indicating greater interclass agreement. Meanwhile, the (+) variant achieves the highest OA (97.22%) and matches the AA (96.92%) of the deeper model, reaffirming its efficient design. These results demonstrate that CFL not only improves classification accuracy in minority or ambiguous land-cover classes but also enhances consistency and generalization, particularly in complex hyperspectral classification settings.

E. Visualization of Attention Weights

To conclude this analysis and with the aim of improving the interpretability of the spectral attention mechanism within the

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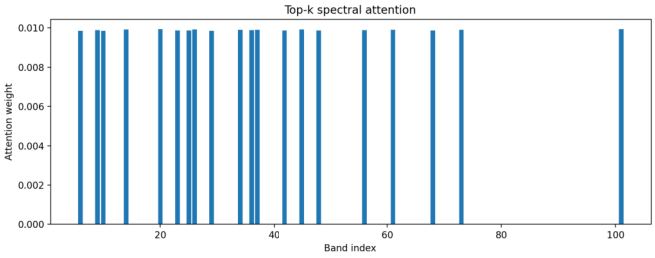


Fig. 13. Visualization of top-20 informative bands of the University of Pavia dataset, which correspond to those channels that exhibit the greatest contribution in terms of attention weight.

proposed method, the spectral attention weights assigned to the spectral bands of the University of Pavia dataset are visualized in Fig. 13. In particular, the top-20 informative bands are reported, which correspond to the channels that exhibit the greatest contribution in terms of attention weight. This discrete representation facilitates the identification of spectral regions that are most relevant to the model, enhancing interpretability.

As observed in Fig. 13, the distribution of selected bands covers all the spectral range rather than being concentrated in a narrow interval, indicating that the discriminative information for land-cover classification in this dataset arises from multiple wavelength regions. This behavior is consistent with the heterogeneity of the University of Pavia scene, which includes different urban materials such as asphalt, bricks, metal roofs, and vegetation. Each of these materials possesses distinctive reflectance characteristics at different spectral intervals, and the attention mechanism effectively isolates the wavelengths that maximize separability among classes.

As a result, the model demonstrates its ability to focus on the most discriminative wavelengths while discarding redundant or noisy information. This contributes not only to a more efficient representation but also to improved generalization, since irrelevant spectral variability is minimized.

V. Conclusion

This work deals with the challenging HSI classification task through the novel `DenseMixerMamba` network. The proposed model introduces an innovative end-to-end solution for HSI classification by tackling the analysis of fine-grained spatial and spectral information in long sequences. In particular, the `DenseMixerMamba` network integrates the latest advances of the current state-of-the-art in DL, i.e., Mamba, attention mechanisms, and dense connectivity architectures. The inclusion of channel attention enables the rapid identification of critical spectral channels, enhancing feature representation. Furthermore, a novel encoder, incorporating the MVM block along with dense connectivity, is developed to facilitate the capture of spatial and spectral patterns without incurring into additional computational overhead. To support this framework, a dual-branch architecture is designed to preserve both spectral and spatial information within the MVM block, ensuring robust final representations.

Extensive experiments conducted on four real-world datasets demonstrate that `DenseMixerMamba` consistently achieves superior classification accuracy and generalization performance compared to state-of-the-art methods. These findings underscore its efficacy and computational efficiency, positioning it as a promising approach for advancing the field of HSI classification. Nevertheless, experimental results show that the performance of the proposed method in few-shot learning is not always satisfactory. In the future, incorporating unsupervised or semi-supervised learning paradigms will further enhance the utility of the model in scenarios where labeled data is scarce, broadening its applicability across diverse real-world use cases.

VI. Acknowledgment

This work made use of the computing infrastructure of CETA-CIEMAT. CETA-CIEMAT belongs to CIEMAT and the Government of Spain.

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